

Polynomial-Logarithmic Transseries

Arithmetic, Composition, Series Reversal,
and the Lambert W Expansion

A detailed tutorial and computational reference

This article develops the concrete transseries algebra

$$\mathbb{K}[\log X]((X^{-1})),$$

its natural field extensions, and its near-identity substitution group. The focus is constructive: every operation is accompanied by a coefficient recursion, a proof of formal validity, a truncation rule, and worked examples. The universal polynomial sequence governing the large-argument expansion of Lambert's W function is derived from formal reversion in three independent ways.

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Abstract

Polynomial-logarithmic transseries are expansions in inverse powers of a large variable whose coefficients are polynomials in its logarithm. The prototype is

$$W(z) = L_1 - L_2 + \sum_{n \geq 1} \frac{P_n(L_2)}{L_1^n}, \quad L_1 = \log z, \quad L_2 = \log \log z,$$

where each P_n is a polynomial of degree n . These expressions are simple enough to manipulate explicitly, but rich enough that ordinary power series intuition is unreliable: X^{-1} is smaller than every fixed inverse power of $\log X$, division may force rational or Laurent-logarithmic coefficients, and composition is meaningful only for substitutions compatible with the scale hierarchy.

We construct the complete filtered ring $\mathcal{R} = \mathbb{K}[\ell][[t]]$, with $t = X^{-1}$ and $\ell = \log X$, and its Laurent version $\mathcal{P} = \mathbb{K}[\ell]((t))$. We give exact rules for multiplication, powers, reciprocals, quotients, logarithms, exponentials, differentiation, and integration. Composition with a near-linear large transseries $G = X + A$, $A \in \mathcal{R}$, is described both by substitution on the generators

$$t \mapsto \frac{t}{1 + tA}, \quad \ell \mapsto \ell + \log(1 + tA),$$

and by the formally summable Taylor series

$$F \circ (X + A) = \sum_{j \geq 0} \frac{A^j}{j!} F^{(j)}(X).$$

The maps $X + A$ form a group under composition. Their inverses are obtained coefficientwise, by a contractive fixed point, by Newton reversion, or by an at-infinity Lagrange–Bürmann formula:

$$(X + \phi)^{[-1]} = X + \sum_{m \geq 1} \frac{(-1)^m}{m!} D_X^{m-1}(\phi^m).$$

For $\phi = \ell$, this yields the Wright omega function and the Lambert polynomials

$$P_n(u) = \frac{(-1)^{n+1}}{(n+1)!} D_u(D_u - 1) \cdots (D_u - n + 1) u^{n+1},$$

including their Stirling-number formula, differential-integral recurrence, generating equation, degree laws, and numerical behavior. The final chapters treat implementation, precision control, extensions to Puiseux and Hahn supports, iterated logarithms, and the boundary between this discrete algebra and full logarithmic-exponential transseries.

How to read this article

The article has three simultaneous goals.

1. It is a tutorial. The scale hierarchy and every formal operation are introduced from first principles.
2. It is a reference. Hypotheses are stated precisely, especially where a frequently written symbolic operation leaves the polynomial-logarithmic ring.
3. It is an algorithmic guide. All calculations are organized around finite truncations, and the final implementation chapter gives tested symbolic code.

A reader mainly interested in Lambert W can read [Chapters 1, 2 and 7 to 9](#) in that order. For a complete algebraic treatment, read sequentially. The symbol X is the large variable used by the formal theory; z is reserved for the argument of Lambert W . Thus, in the Lambert application, $X = \log z$.

Three meanings of “series”

Throughout, a displayed infinite expansion can have three logically distinct statuses.

1. It may be a *formal* identity in a complete filtered ring.
2. It may be an *asymptotic* expansion of an analytic function.
3. It may be a genuinely *convergent* numerical series.

Formal algebra alone proves the first. Additional analytic information proves the second or third. For the classical large-argument Lambert series, all three viewpoints are available on appropriate domains, but they should not be silently identified.

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Part I

The scale and its algebra

Chapter 1

Lambert W as the motivating inverse problem

1.1 Why a logarithm of a logarithm appears

Lambert's function is defined implicitly by

$$W(z)e^{W(z)} = z. \quad (1.1)$$

On the positive real axis, $W(z) \rightarrow +\infty$ with z , so taking logarithms is legitimate and gives

$$W(z) + \log W(z) = \log z. \quad (1.2)$$

Put

$$L_1 = \log z.$$

The second term in Equation (1.2) is smaller than the first, so the dominant balance is $W \sim L_1$. Feeding this approximation back into the smaller term gives

$$\log W \sim \log L_1 =: L_2, \quad W \sim L_1 - L_2.$$

A third substitution reveals that the next correction is of order L_2/L_1 . Continuing produces

$$W(z) = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2^2/2 - L_2}{L_1^2} + \frac{L_2^3/3 - 3L_2^2/2 + L_2}{L_1^3} + \dots. \quad (1.3)$$

The structural feature is more important than these first coefficients:

$$W(z) \in L_1 - L_2 + L_1^{-1} \mathbb{K}[L_2][[L_1^{-1}]]. \quad (1.4)$$

Every inverse power of the large scale L_1 carries a finite polynomial in the slower scale $L_2 = \log L_1$.

1.2 Why an ordinary Laurent series is insufficient

An ordinary Laurent series in L_1^{-1} has constant coefficients. It cannot contain even the first correction $-L_2$, much less the growing polynomials $P_n(L_2)$. Treating L_1^{-1} and L_2 as two independent small variables is also wrong: as $z \rightarrow \infty$,

$$L_1^{-1} \rightarrow 0, \quad L_2 \rightarrow \infty,$$

and their relative size is rigid:

$$L_1^{-1}L_2^a = o(L_2^{-b}) \quad \text{for every fixed } a, b \in \mathbb{R}. \quad (1.5)$$

In words, one factor of L_1^{-1} beats every fixed power of L_2 . This is the first genuinely transserial feature of the calculation.

1.3 The reduced model

To isolate the reusable algebra, forget temporarily that $X = \log z$. Let

$$X \rightarrow +\infty, \quad t = X^{-1}, \quad \ell = \log X.$$

The basic objects will be

$$A(X) = \sum_{n \geq n_0} P_n(\ell)t^n, \quad P_n \in \mathbb{K}[\ell], \quad (1.6)$$

with only finitely many negative n . The monomials are

$$t^n \ell^k = X^{-n}(\log X)^k.$$

The Lambert expansion is recovered by setting $X = L_1$, $\ell = L_2$.

1.4 Four operations, four different issues

The requested operations are superficially familiar, but each hides a different structural question.

Multiplication.

The Cauchy product is always defined because only finitely many pairs of t -orders contribute to a fixed coefficient.

Division.

A nonzero leading coefficient need not be invertible in $\mathbb{K}[\ell]$. Division by $\ell + t$, for example, forces inverse powers of ℓ . One must decide whether to remain in a ring or enlarge to a field.

Composition.

The expression $A \circ G$ is not defined for arbitrary formal objects. The inner series must tend to the asymptotic region of the outer one, and its leading scale determines whether new monomials such as $\log \ell$ or fractional powers of t appear.

Series reversal.

Compositional inversion near infinity uses a different filtration from ordinary reversion at the origin. Its key fact is that differentiation raises the t -order, making fixed-point and Lagrange expansions formally summable.

The rest of the article turns these observations into exact theorems and algorithms.

Chapter 2

The polynomial-logarithmic ring

2.1 Coefficient field and basic rings

Let $\mathbb{K}$ be a field of characteristic zero, usually $\mathbb{R}$ or $\mathbb{C}$. Introduce two formal symbols t and ℓ , later interpreted as X^{-1} and $\log X$.

Definition 2.1 (Polynomial-logarithmic series ring). *The ring of polynomial-logarithmic series is*

$$\mathcal{R} := \mathbb{K}[\ell][[t]]. \quad (2.1)$$

Its Laurent extension in the fast small parameter is

$$\mathcal{P} := \mathbb{K}[\ell]((t)) = \bigcup_{m \geq 0} t^{-m} \mathcal{R}. \quad (2.2)$$

Thus every $A \in \mathcal{P}$ has a unique representation

$$A = \sum_{n=n_0}^{\infty} P_n(\ell) t^n, \quad P_n \in \mathbb{K}[\ell], \quad (2.3)$$

for some integer n_0 .

The adjective “polynomial-logarithmic” refers to the coefficient blocks $P_n(\ell)$. It does not mean that the whole expression is a polynomial in two variables: infinitely many t -orders are allowed, and the variables carry an asymptotic hierarchy.

2.2 Monomial order and dominance

Write

$$P_n(\ell) = \sum_{k=0}^{d_n} a_{n,k} \ell^k.$$

The support of A is

$$\text{supp } A = \{(n, k) \in \mathbb{Z} \times \mathbb{N} : a_{n,k} \neq 0\}.$$

At $X \rightarrow +\infty$, monomials are ordered by

$$t^n \ell^k \succ t^m \ell^j \iff (n < m) \text{ or } (n = m \text{ and } k > j). \quad (2.4)$$

Indeed, if $n < m$, then

$$\frac{t^m \ell^j}{t^n \ell^k} = X^{-(m-n)} (\log X)^{j-k} \longrightarrow 0$$

regardless of the fixed log powers. Within one t -block, ordinary polynomial degree determines the leading monomial.

Example 2.2. *The monomials*

$$X^2, \quad X(\log X)^{100}, \quad X, \quad (\log X)^7, \quad 1, \quad X^{-1}(\log X)^{1000}$$

are listed from largest to smallest. No finite log power can compensate for one lost power of X .

2.3 The t -adic valuation

For nonzero A as in Equation (2.3), define

$$v_t(A) = \min\{n : P_n \neq 0\}, \quad v_t(0) = +\infty. \quad (2.5)$$

Then

$$v_t(AB) = v_t(A) + v_t(B), \quad (2.6)$$

$$v_t(A + B) \geq \min\{v_t(A), v_t(B)\}. \quad (2.7)$$

The valuation sees the fast scale t but deliberately ignores the degree in ℓ . This coarser filtration is exactly what makes infinite algebraic operations manageable.

For $N \in \mathbb{Z}$, write

$$A = B + O_t(t^N) \iff v_t(A - B) \geq N.$$

The truncation below order N is

$$[A]_{<N} := \sum_{n < N} P_n(\ell) t^n. \quad (2.8)$$

For a fixed Laurent lower bound, this is a finite sum.

2.4 Completeness and coefficient stabilization

A sequence (A_j) converges t -adically to A if

$$v_t(A_j - A) \longrightarrow +\infty.$$

Equivalently, for each fixed n , the polynomial coefficient of t^n eventually stabilizes. The ring $\mathcal{R}$ and the Laurent ring $\mathcal{P}$ are complete in this sense.

Proposition 2.3 (Formal summability criterion). *Let $(S_j)_{j \geq 0} \subset \mathcal{P}$. If $v_t(S_j) \rightarrow +\infty$, then the sum $\sum_{j \geq 0} S_j$ is well defined in $\mathcal{P}$: for every N , only finitely many S_j affect the truncation below t^N .*

Proof. Choose J such that $v_t(S_j) \geq N$ for $j \geq J$. Then $[\sum_j S_j]_{<N} = [\sum_{j < J} S_j]_{<N}$, so each finite truncation is unambiguous. Compatibility of these truncations as N varies defines a unique element of the complete ring. $\square$

This criterion replaces numerical convergence throughout the formal theory.

2.5 Asymptotic realization

Suppose an actual function $f(X)$ is defined for sufficiently large positive X . A useful blockwise meaning of

$$f(X) \sim \sum_{n=n_0}^{\infty} P_n(\log X) X^{-n}$$

is that for every integer $N > n_0$, there exists a finite exponent M_N such that

$$f(X) - \sum_{n=n_0}^{N-1} P_n(\log X) X^{-n} = O\left(X^{-N} (\log X)^{M_N}\right). \quad (2.9)$$

A sharper monomial-by-monomial expansion can be obtained by ordering every $X^{-n}(\log X)^k$ separately as in [Equation \(2.4\)](#). The blockwise form is usually more convenient computationally.

Remark 2.4. *Formal identities in $\mathcal{R}$ do not assert that substituting a numerical value for X makes the infinite sum converge. They assert that every finite coefficient block follows from algebra compatible with the asymptotic scale. When the series is an asymptotic expansion of a function, finite truncations have the usual remainder interpretation. When it is convergent, the formal sum also has a numerical value.*

2.6 Why $\mathcal{P}$ is not a field

The coefficient ring $\mathbb{K}[\ell]$ is not a field. Consequently $\mathbb{K}[\ell]((t))$ is not a field either. For example, $\ell + t$ is nonzero, but its reciprocal begins

$$\frac{1}{\ell + t} = \ell^{-1} - \ell^{-2}t + \ell^{-3}t^2 - \dots, \quad (2.10)$$

and the coefficients are not polynomials in ℓ .

Two useful enlargements are

$$\mathbb{K}(\ell)((t)) \quad (\text{rational functions of } \ell \text{ as coefficients}), \quad (2.11)$$

$$\mathcal{F} := \mathbb{K}((\ell^{-1}))((t)) \quad (\text{Laurent series at } \ell = \infty \text{ as coefficients}). \quad (2.12)$$

Both are fields. The second records the refined hierarchy

$$t \prec \ell^{-m} \prec 1 \prec \ell^m \prec t^{-1} \quad (m > 0).$$

The distinction between $\mathcal{P}$ and these field extensions will matter in [Chapter 5](#).

2.7 Examples and nonexamples

Expression	Membership and reason
$3\ell^2 - 1 + \sum_{n \geq 1} (\ell^n + 1)t^n$	In $\mathcal{R}$; every t -coefficient is a finite polynomial.
$t^{-3}(\ell + 1) + \sum_{n \geq 0} \ell^{2n}t^n$	In $\mathcal{P}$; only finitely many negative t -powers occur.
$\sum_{k \geq 0} \ell^{-k}$	In $\mathbb{K}((\ell^{-1}))$, not in $\mathbb{K}[\ell]$.
$\log \ell$	Not in the one-logarithm ring; it requires the next logarithmic level.
$\sum_{q \geq 0} t^{q/2} \ell^q$	A Puiseux-logarithmic series, not an element of $\mathcal{P}$.
e^{-X}	Exponentially small; it belongs to a larger transseries field, not this power-log level.

Core viewpoint

The ring $\mathcal{R} = \mathbb{K}[\ell][[t]]$ is a complete t -adically filtered ring. A calculation through order t^N is finite. Polynomial dependence on ℓ is retained exactly inside each block, while all infinite processes must push their new contributions to successively higher powers of t .

Chapter 3

Differential calculus on the scale

3.1 The derivation in (t, ℓ) coordinates

Under the interpretation $t = X^{-1}$, $\ell = \log X$,

$$\frac{dt}{dX} = -t^2, \quad \frac{d\ell}{dX} = t.$$

Therefore the formal derivation is

$$D_X = t \partial_\ell - t^2 \partial_t. \quad (3.1)$$

Applied to one coefficient block,

$$D_X(t^n P(\ell)) = t^{n+1} (P'(\ell) - nP(\ell)). \quad (3.2)$$

It is useful to introduce $D = D_\ell$. Then repeated differentiation has a closed form.

Proposition 3.1 (Repeated derivative). *For $r \geq 0$,*

$$D_X^r(t^n P(\ell)) = t^{n+r} \prod_{j=0}^{r-1} (D - n - j) P(\ell), \quad (3.3)$$

where the empty product for $r = 0$ is the identity.

Proof. The case $r = 1$ is [Equation \(3.2\)](#). If the formula holds at r , differentiating the factor $t^{n+r} Q(\ell)$ applies $D - (n + r)$ to Q , which appends the next factor. All factors are polynomials in D , so their order is immaterial. $\square$

In particular,

$$D_X^r P(\ell) = t^r D(D - 1) \cdots (D - r + 1) P(\ell). \quad (3.4)$$

The falling differential operator in [Equation \(3.4\)](#) is the source of the Stirling numbers in the Lambert expansion.

3.2 Differentiation raises fast order

From [Equation \(3.2\)](#),

$$\nu_t(D_X A) \geq \nu_t(A) + 1 \quad (3.5)$$

for every $A \in \mathcal{P}$, with strict inequality possible if the leading block is annihilated. This elementary fact powers the composition and reversion theory: a Taylor term containing the r -th derivative automatically gains at least r powers of t .

The Euler derivation $XD_X = t^{-1}D_X$ preserves t -order and acts on a block by

$$XD_X(t^n P(\ell)) = t^n (D - n)P(\ell). \quad (3.6)$$

3.3 Formal integration

The same algebra is closed under antiderivatives. To integrate $t^n P(\ell)$, seek $t^{n-1} Q(\ell)$. Equation Equation (3.2) gives

$$Q' - (n - 1)Q = P. \quad (3.7)$$

If $n \neq 1$, the operator $D - (n - 1)$ is invertible on polynomials and

$$Q = - \sum_{j=0}^{\deg P} \frac{P^{(j)}}{(n - 1)^{j+1}}. \quad (3.8)$$

The sum terminates because high derivatives vanish. If $n = 1$, then $Q' = P$, so any polynomial antiderivative of P works. Thus

$$\int \mathcal{P} \, dX \subseteq \mathcal{P} + \mathbb{K}. \quad (3.9)$$

Example 3.2. Using Equation (3.8),

$$\int X^{-3} (\log X)^2 \, dX = -\frac{X^{-2}}{2} \left((\log X)^2 + \log X + \frac{1}{2} \right) + C.$$

Differentiation verifies the result immediately.

3.4 Small-series functional calculus

Let $U \in t\mathcal{R}$, so $v_t(U) \geq 1$. By Theorem 2.3, the following standard formal series are well defined:

$$(1 + U)^\alpha = \sum_{m \geq 0} \binom{\alpha}{m} U^m, \quad (3.10)$$

$$\log(1 + U) = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} U^m, \quad (3.11)$$

$$\exp(U) = \sum_{m \geq 0} \frac{U^m}{m!}. \quad (3.12)$$

For a target precision t^N , only $m < N$ can contribute. These formulas will repeatedly appear in reciprocals and composition.

Warning 3.3. *The smallness hypothesis is not decorative. The formal series for $\log(1 + U)$ is not t -adically summable if $v_t(U) = 0$. A non-small leading term must first be factored or absorbed into the coefficient field.*

Part II

Arithmetic

Chapter 4

Addition, multiplication, powers, and elementary functions

4.1 Coefficientwise addition

If

$$A = \sum_n P_n(\ell)t^n, \quad B = \sum_n Q_n(\ell)t^n,$$

then

$$A + B = \sum_n (P_n + Q_n)t^n. \quad (4.1)$$

Nothing subtle occurs except cancellation: the valuation of a sum can exceed the smaller input valuation when leading coefficient blocks cancel.

4.2 The Cauchy product

Multiplication is the familiar convolution in the fast index.

Theorem 4.1 (Cauchy multiplication). *For $A, B \in \mathcal{P}$,*

$$AB = \sum_n R_n(\ell)t^n, \quad R_n(\ell) = \sum_{j+k=n} P_j(\ell)Q_k(\ell). \quad (4.2)$$

For each fixed n , the inner sum is finite.

Proof. Both Laurent supports are bounded below, say by j_0 and k_0 . If $j + k = n$, then $j_0 \leq j \leq n - k_0$, so only finitely many pairs occur. Each product $P_j Q_k$ is a polynomial in ℓ . Hence the coefficient R_n is well defined in $\mathbb{K}[\ell]$, and the product remains in $\mathcal{P}$. $\square$

If $d_j = \deg P_j$ and $e_k = \deg Q_k$, then

$$\deg R_n \leq \max_{j+k=n} (d_j + e_k). \quad (4.3)$$

This elementary bound is useful for preallocating polynomial storage.

4.3 A complete multiplication example

Let

$$\begin{aligned} A &= (1 + 2\ell) + (\ell^2 - 1)t + 3\ell t^2 + (2 - \ell)t^3, \\ B &= 2 - \ell t + (\ell + 1)t^2 + \ell^2 t^3. \end{aligned}$$

The first four coefficient convolutions are

$$\begin{aligned} R_0 &= 2(1 + 2\ell) = 2 + 4\ell, \\ R_1 &= (1 + 2\ell)(-\ell) + 2(\ell^2 - 1) = -\ell - 2, \\ R_2 &= (1 + 2\ell)(\ell + 1) + (\ell^2 - 1)(-\ell) + 6\ell \\ &= -\ell^3 + 2\ell^2 + 10\ell + 1, \\ R_3 &= (1 + 2\ell)\ell^2 + (\ell^2 - 1)(\ell + 1) + (3\ell)(-\ell) + 2(2 - \ell) \\ &= 3\ell^3 - \ell^2 - 3\ell + 3. \end{aligned}$$

Thus

$$\begin{aligned} AB &= (2 + 4\ell) + (-\ell - 2)t + (-\ell^3 + 2\ell^2 + 10\ell + 1)t^2 \\ &\quad + (3\ell^3 - \ell^2 - 3\ell + 3)t^3 + O_t(t^4). \end{aligned} \tag{4.4}$$

The calculation is ordinary polynomial arithmetic inside each block and ordinary series convolution across blocks.

4.4 Truncated multiplication

Suppose $A, B \in \mathcal{R}$ and only terms below t^N are needed. Then

$$[AB]_{<N} = [[A]_{<N} [B]_{<N}]_{<N}. \tag{4.5}$$

No discarded coefficient can return to a lower order, because valuations add. The direct algorithm is therefore:

Algorithm 4.2 (Truncated Cauchy product). *Store $A = (P_0, \dots, P_{N-1})$ and $B = (Q_0, \dots, Q_{N-1})$. Initialize $R_n = 0$. For every pair $j, k \geq 0$ with $j + k < N$, accumulate $P_j Q_k$ into R_{j+k} .*

The naive cost is $O(N^2)$ polynomial multiplications. Fast convolution can reduce the dependence on N , but for logarithmic coefficient degrees that grow with N , polynomial arithmetic and expression swell often dominate before FFT methods become useful.

4.5 Integer powers

For $m \geq 0$,

$$[t^n]A^m = \sum_{j_1 + \dots + j_m = n} P_{j_1} \cdots P_{j_m}. \tag{4.6}$$

If $A \in t\mathcal{R}$, every $j_r \geq 1$, so the sum vanishes for $m > n$. This finiteness is the reason a formal function of a small series is easy to compute.

Repeated squaring is preferable when one fixed integer power is required. For a whole family of powers, as in $\log(1 + A)$, it is often cheaper to update successively:

$$A^0 = 1, \quad A^{m+1} = A^m A,$$

truncating after each multiplication.

4.6 Generalized binomial powers

If $U \in t\mathcal{R}$ and $\alpha \in \mathbb{K}$, then

$$(1 + U)^\alpha = 1 + \alpha U + \frac{\alpha(\alpha - 1)}{2} U^2 + \dots$$

is defined by Equation (3.10). More generally, if

$$A = a t^r (1 + U), \quad a \in \mathbb{K}^\times, \quad r \in \mathbb{Z}, \quad U \in t\mathcal{R},$$

then an integer power stays in $\mathcal{P}$, while a noninteger power has the formal form

$$A^\alpha = a^\alpha t^{\alpha r} (1 + U)^\alpha. \quad (4.7)$$

Unless $\alpha r \in \mathbb{Z}$, this leaves $\mathcal{P}$ and belongs to a Puiseux or Hahn extension. Even when αr is integral, a branch for a^α must be selected over $\mathbb{C}$.

4.7 Logarithm and exponential

A logarithm is defined after factoring the dominant monomial. For

$$A = a t^r (1 + U), \quad U \in t\mathcal{R},$$

we obtain

$$\log A = \log a + r \log t + \log(1 + U) = \log a - r\ell + \log(1 + U). \quad (4.8)$$

Thus the logarithm remains polynomial-logarithmic when the leading coefficient is a nonzero scalar. If the leading coefficient is a nonconstant polynomial $P(\ell)$, then $\log P(\ell)$ generally introduces $\log \ell$, a new scale.

The exponential behaves differently. If $U \in t\mathcal{R}$, then $\exp U \in 1 + t\mathcal{R}$. But $\exp(\ell) = X = t^{-1}$, while $\exp(\ell^2)$ is neither a power of t nor a polynomial-logarithmic series. Full closure under exponentiation requires a logarithmic-exponential transseries field.

Example 4.3 (A finite logarithm calculation). *Let*

$$U = (\ell + 1)t + \ell^2 t^2.$$

Since $U^4 = O_t(t^4)$,

$$\begin{aligned} \log(1 + U) &= U - \frac{U^2}{2} + \frac{U^3}{3} + O_t(t^4) \\ &= (\ell + 1)t + \left(\frac{\ell^2}{2} - \ell - \frac{1}{2} \right) t^2 \\ &\quad + \left(-\ell^3 - \ell^2 + \frac{(\ell + 1)^3}{3} \right) t^3 + O_t(t^4). \end{aligned}$$

Every operation was finite after the target order was fixed.

4.8 Formal substitution into a scalar power series

Let $f(u) = \sum_{m \geq 0} c_m u^m \in \mathbb{K}[[u]]$ and let $U \in t\mathcal{R}$. Define

$$f(U) := \sum_{m \geq 0} c_m U^m. \quad (4.9)$$

The sum is formally valid because $\nu_t(U^m) \geq m$. If one needs terms below t^N , only $c_0, \dots, c_{N-1}$ matter. This is the basic engine for binomial powers, reciprocals, logarithms, exponentials of small corrections, and the generator formulas for composition.

Arithmetic invariant

In every valid infinite arithmetic construction, the m -th summand must have t -valuation tending to infinity. Once that condition is checked, all coefficient formulas reduce to finite polynomial arithmetic.

Chapter 5

Division and reciprocal series

5.1 Units in the polynomial-logarithmic ring

The first question is not how to divide, but where the quotient is expected to live.

Theorem 5.1 (Unit criterion). *Let*

$$B = t^r (B_0(\ell) + B_1(\ell)t + B_2(\ell)t^2 + \cdots) \in \mathcal{P}, \quad B_0 \neq 0.$$

Then B is invertible in $\mathcal{P}$ if and only if $B_0 \in \mathbb{K}^\times$, that is, the leading coefficient block is a nonzero constant.

Proof. The factor t^r is invertible in $\mathcal{P}$. If $B_0 = b \in \mathbb{K}^\times$, write

$$B = t^r b(1 + U), \quad U \in t\mathcal{R}.$$

Then

$$B^{-1} = t^{-r} b^{-1} \sum_{m \geq 0} (-U)^m \in \mathcal{P}.$$

Conversely, suppose $BC = 1$ in $\mathcal{P}$. After matching the Laurent powers of t , the product of the leading coefficient blocks of B and C must equal 1 in $\mathbb{K}[\ell]$. The only units of a polynomial ring over a field are nonzero constants, so $B_0 \in \mathbb{K}^\times$. $\square$

5.2 Coefficient recursion for a quotient

Assume first that

$$\begin{aligned} A &= \sum_{n \geq 0} A_n(\ell) t^n, \\ B &= b + \sum_{n \geq 1} B_n(\ell) t^n, \quad b \in \mathbb{K}^\times, \end{aligned}$$

and seek $C = A/B = \sum_{n \geq 0} C_n(\ell) t^n$. Equating coefficients in $BC = A$ gives

$$C_n = b^{-1} \left(A_n - \sum_{j=1}^n B_j C_{n-j} \right), \quad n \geq 0, \tag{5.1}$$

where the sum is empty at $n = 0$. This is formal long division.

Proposition 5.2. *Recurrence Equation (5.1) determines a unique quotient $C \in \mathcal{R}$. If input coefficients through order N are known, then the quotient through order N requires no higher input coefficients.*

Proof. At stage n , only $C_0, \dots, C_{n-1}$ occur on the right. Since b is invertible in $\mathbb{K}$, C_n is uniquely determined and is polynomial in ℓ . The second assertion is immediate from the same triangular structure. $\square$

Laurent lower bounds are handled by factoring powers of t from numerator and denominator.

5.3 Worked reciprocal example

Consider

$$B = 1 + (\ell + 1)t + \ell^2 t^2 + (\ell - 2)t^3. \quad (5.2)$$

Put $C = B^{-1} = \sum C_n t^n$. Since $A = 1$ in Equation (5.1),

$$\begin{aligned} C_0 &= 1, \\ C_1 &= -(\ell + 1), \\ C_2 &= -(\ell + 1)C_1 - \ell^2 C_0 = 2\ell + 1, \\ C_3 &= -(\ell + 1)C_2 - \ell^2 C_1 - (\ell - 2)C_0 \\ &= \ell^3 - \ell^2 - 4\ell + 1. \end{aligned}$$

Continuing,

$$\begin{aligned} B^{-1} &= 1 - (\ell + 1)t + (2\ell + 1)t^2 + (\ell^3 - \ell^2 - 4\ell + 1)t^3 \\ &\quad + (-\ell^4 - 2\ell^3 + 5\ell^2 + 2\ell - 3)t^4 + O_t(t^5). \end{aligned} \quad (5.3)$$

Multiplying Equation (5.2) and Equation (5.3) is the simplest independent check.

5.4 Geometric inversion

The proof of Theorem 5.1 gives another algorithm. With

$$B = b(1 + U), \quad U \in t\mathcal{R},$$

we have

$$B^{-1} = b^{-1}(1 - U + U^2 - U^3 + \dots). \quad (5.4)$$

At precision t^N , truncate after U^{N-1} . The coefficient recursion is usually more economical for one reciprocal; geometric inversion is useful when powers of U are already available for other purposes.

5.5 Newton inversion

For high precision, Newton iteration replaces linear convergence in the number of correct blocks by quadratic convergence. If V approximates B^{-1} , set

$$V_{\text{new}} = V(2 - BV). \quad (5.5)$$

Then

$$1 - BV_{\text{new}} = (1 - BV)^2. \quad (5.6)$$

Thus, if $BV = 1 + O_t(t^m)$, then $BV_{\text{new}} = 1 + O_t(t^{2m})$. Each step is performed only to the new target precision, for example 1, 2, 4, 8, $\dots$ blocks.

Algorithm 5.3 (Newton reciprocal to order N). *Let $b = B_0 \in \mathbb{K}^\times$ and initialize $V = b^{-1}$. While the known precision $m < N$, replace*

$$V \leftarrow [V(2 - BV)]_{<\min(2m, N)}$$

and double m . Return V .

With fast multiplication, Newton inversion has essentially the same asymptotic complexity as multiplication. Even with naive multiplication, it is often faster than coefficient recursion for large N , because most work occurs in the last doubling step.

5.6 Division by a nonunit

If $B_0(\ell)$ is nonconstant, three cases must be distinguished.

5.6.1 A particular quotient may still lie in $\mathcal{P}$

The equation $BC = A$ can be solved coefficientwise, but at each stage the right-hand polynomial must be divisible by B_0 in $\mathbb{K}[\ell]$. Exact cancellation can therefore make a quotient exist even though B has no global reciprocal.

5.6.2 Exact division in a rational coefficient field

In $\mathbb{K}(\ell)((t))$, every nonzero $B_0(\ell)$ is invertible, so Equation (5.1) works with rational functions. For example,

$$\frac{1}{\ell + t} = \frac{1}{\ell} - \frac{t}{\ell^2} + \frac{t^2}{\ell^3} - \cdots \in \mathbb{K}(\ell)[[t]]. \quad (5.7)$$

5.6.3 Asymptotic expansion inside the log scale

A rational coefficient can be expanded at $\ell = \infty$. For instance,

$$\frac{1}{\ell + 1} = \ell^{-1} (1 + \ell^{-1})^{-1} = \ell^{-1} - \ell^{-2} + \ell^{-3} - \cdots. \quad (5.8)$$

This lives in $\mathbb{K}((\ell^{-1}))$, and the full quotient belongs to the iterated Laurent field $\mathcal{F} = \mathbb{K}((\ell^{-1}))((t))$.

Do not hide a field extension

Writing $(\ell + t)^{-1}$ and then manipulating it as though it belonged to $\mathbb{K}[\ell]((t))$ silently changes the algebra. A rigorous calculation should say whether coefficients are polynomial, rational in ℓ , or Laurent series in ℓ^{-1} . This choice affects closure, normalization, and the meaning of truncation.

5.7 Quotients with dominant monomials

Suppose

$$A = t^a A_0(\ell)(1 + U), \quad B = t^b B_0(\ell)(1 + V), \quad U, V \in t\mathcal{R}.$$

In a coefficient field containing A_0/B_0 ,

$$\frac{A}{B} = t^{a-b} \frac{A_0}{B_0} (1 + U)(1 + V)^{-1}. \quad (5.9)$$

This “factor first, expand second” rule prevents invalid geometric expansions. The small object is V , not the entire denominator.

5.8 Residual verification

A truncated quotient $C_{<N}$ is correct through order t^{N-1} exactly when

$$BC_{<N} - A = O_t(t^N). \quad (5.10)$$

Residual checks are cheap, independent of the method used to compute C , and should be part of every symbolic implementation.

Division decision

Before dividing, inspect the leading t -coefficient of the denominator. A nonzero scalar gives a reciprocal in $\mathcal{P}$. A nonconstant polynomial usually forces $\mathbb{K}(\ell)((t))$ or the asymptotically finer field $\mathbb{K}((\ell^{-1}))((t))$. Once the coefficient domain is fixed, quotient coefficients follow from a triangular recursion or Newton inversion.

Part III

Composition

Chapter 6

Composition at infinity

6.1 Why composition needs hypotheses

For convergent functions, $F \circ G$ means evaluating F at the value $G(X)$. For formal asymptotic expansions, neither “evaluation” nor arbitrary substitution is automatically meaningful. Three questions must be answered.

1. Does $G(X)$ tend to the asymptotic region in which F is expanded? For the present article, that means $G(X) \rightarrow +\infty$, or an explicitly chosen complex sector.
2. Can the basic generators G^{-1} and $\log G$ be expanded in the chosen monomial scale?
3. For each requested output order, do only finitely many input terms contribute?

The restricted algebra gives especially clean answers for substitutions asymptotic to a positive multiple of a power of X , and the near-identity case is closed under inversion.

6.2 Composition is determined on the generators

Let

$$G(X) = cX^\rho(1+h), \quad c > 0, \quad \rho > 0, \quad h \prec 1. \quad (6.1)$$

In the discrete ring, take $\rho \in \mathbb{N}_{>0}$ and $h \in t\mathcal{R}$. Then

$$t \circ G = G^{-1} = c^{-1}t^\rho(1+h)^{-1}, \quad (6.2)$$

$$\ell \circ G = \log G = \log c + \rho\ell + \log(1+h). \quad (6.3)$$

The reciprocal and logarithm on the right are defined by small-series expansions. Since every $F \in \mathcal{P}$ is assembled from t and ℓ , these two formulas determine composition.

Definition 6.1 (Substitution homomorphism). *For G as in Equation (6.1), define*

$$\Phi_G(F) := F \circ G = \sum_{n=n_0}^{\infty} P_n(\log c + \rho\ell + \lambda) c^{-n} t^{\rho n} (1+h)^{-n}, \quad \lambda = \log(1+h). \quad (6.4)$$

All factors are expanded t -adically.

Because P_n is a polynomial, the shift by the small series λ is finite:

$$P_n(a + \lambda) = \sum_{j=0}^{\deg P_n} \frac{P_n^{(j)}(a)}{j!} \lambda^j, \quad a = \log c + \rho\ell. \quad (6.5)$$

No analytic Taylor convergence is involved.

Theorem 6.2 (Closure under monomial-leading composition). *If $F \in \mathcal{P}$, $\rho \in \mathbb{N}_{>0}$, and $G = cX^\rho(1 + h)$ with $h \in t\mathcal{R}$, then $F \circ G \in \mathcal{P}$. The map $\Phi_G : \mathcal{P} \rightarrow \mathcal{P}$ is a $\mathbb{K}$ -algebra homomorphism and is continuous for the t -adic topology.*

Proof. For one block $P_n(\ell)t^n$, equations (6.2) and (6.3) give a leading power $t^{\rho n}$ multiplied by series in t with polynomial coefficients. The Laurent lower bound remains finite because the input has only finitely many negative n . At a fixed output order N , only finitely many input blocks with $\rho n < N$ contribute, and only finitely many powers in each binomial, logarithmic, and polynomial-shift expansion contribute. Thus the sum is well-defined in $\mathcal{P}$. Substitution on the generators respects sums and products, so it is an algebra homomorphism. Finally, a term of order t^n maps to order at least $t^{\rho n}$ up to a fixed Laurent shift, proving continuity. $\square$

Remark 6.3. *For rational $\rho = p/q > 0$, the same formula is valid after adjoining $t^{1/q}$. For arbitrary real ρ , a Hahn-series exponent group is the natural setting. The algebraic rule does not change; only the support domain does.*

6.3 The near-linear case

The most important substitution for reversion is

$$G = X + A, \quad A \in \mathcal{R}. \quad (6.6)$$

Since every polynomial in $\log X$ is $o(X)$, this is asymptotic to X . Factor

$$G = X(1 + tA).$$

With $h = tA \in t\mathcal{R}$, equations (6.2) and (6.3) specialize to

$$t \circ (X + A) = \frac{t}{1 + tA}, \quad (6.7)$$

$$\ell \circ (X + A) = \ell + \log(1 + tA). \quad (6.8)$$

Therefore

$$F \circ (X + A) = \sum_n P_n(\ell + \log(1 + tA)) t^n (1 + tA)^{-n}. \quad (6.9)$$

The central substitution rule

For a near-identity change of the large variable, never substitute $t \mapsto t + A$. The variable t means $1/X$, so the correct rule is

$$t \mapsto \frac{t}{1 + tA}.$$

At the same time, the logarithmic coordinate changes by

$$\ell \mapsto \ell + \log(1 + tA).$$

Both changes are essential.

6.4 A worked composition

Take

$$F = \ell^2 + (\ell + 1)t + (\ell^2 - \ell)t^2 + (2\ell + 1)t^3. \quad (6.10)$$

Let

$$G = X + \ell, \quad h = t\ell.$$

The generator substitutions through order t^3 are

$$\begin{aligned} t \circ G &= t - \ell t^2 + \ell^2 t^3 + O_t(t^4), \\ \ell \circ G &= \ell + \ell t - \frac{\ell^2}{2} t^2 + \frac{\ell^3}{3} t^3 + O_t(t^4). \end{aligned}$$

Substituting simultaneously and collecting blocks gives

$$\begin{aligned} F \circ G &= \ell^2 + (2\ell^2 + \ell + 1)t + (-\ell^3 + \ell^2 - \ell)t^2 \\ &\quad + \left(\frac{2}{3}\ell^4 - 2\ell^3 + \frac{7}{2}\ell^2 + \ell + 1 \right) t^3 + O_t(t^4). \end{aligned} \quad (6.11)$$

A sequential substitution can be wrong because the replacement for ℓ contains t , and the replacement for t contains ℓ . Software should substitute the two generators simultaneously or use the Taylor method below.

6.5 Formal Taylor composition

The near-linear case has a second, remarkably simple description.

Theorem 6.4 (Taylor theorem at infinity). *For $F \in \mathcal{P}$ and $A \in \mathcal{R}$,*

$$F \circ (X + A) = \sum_{j=0}^{\infty} \frac{A^j}{j!} D_X^j F. \quad (6.12)$$

The series is t -adically summable. To determine the result below order t^N , only finitely many derivatives are needed.

Proof. By Equation (3.5),

$$\nu_t(A^j D_X^j F) \geq \nu_t(F) + j + j\nu_t(A) \geq \nu_t(F) + j,$$

because $A \in \mathcal{R}$ has nonnegative valuation. Hence the right side is summable by Theorem 2.3.

It remains to identify the sum with generator substitution. For $F = t = X^{-1}$,

$$D_X^j t = (-1)^j j! t^{j+1}.$$

After this, Equation (6.12) becomes

$$t \sum_{j \geq 0} (-tA)^j = \frac{t}{1 + tA},$$

which is Equation (6.7). For $F = \ell$,

$$D_X^j \ell = (-1)^{j-1} (j-1)! t^j \quad (j \geq 1),$$

so Taylor's formula becomes

$$\ell + \sum_{j \geq 1} \frac{(-1)^{j-1}}{j} (tA)^j = \ell + \log(1 + tA),$$

which is Equation (6.8). Both sides of Equation (6.12) are continuous algebra homomorphisms in F and agree on the generators, hence agree on all of $\mathcal{P}$. $\square$

Corollary 6.5 (First variation). *For $A \in \mathcal{R}$,*

$$F \circ (X + A) = F + AF' + \frac{A^2}{2} F'' + \cdots. \quad (6.13)$$

If only the first new t -block is needed, the linear term often suffices because every derivative raises the fast order.

6.6 Precision planning for Taylor composition

Suppose $F \in \mathcal{R}$ and the result is required modulo t^N . Since $v_t(D_X^j F) \geq j$, all terms with $j \geq N$ vanish modulo t^N . Thus

$$F \circ (X + A) = \sum_{j=0}^{N-1} \frac{A^j}{j!} D_X^j F + O_t(t^N). \quad (6.14)$$

If F begins at t^r , only $j < N - r$ are needed. Every derivative and product should be truncated immediately; carrying higher blocks causes large unnecessary intermediate polynomials.

Algorithm 6.6 (Taylor composition to order N). *Initialize $S = 0$, $D_0 = F$, and $A_0 = 1$. For $j = 0, 1, \dots, N-1$: add $A_j D_j / j!$ to S , truncate S , update*

$$A_{j+1} = [A_j A]_{<N}, \quad D_{j+1} = [D_X D_j]_{<N}.$$

Stop early when $v_t(D_j) \geq N$.

6.7 Composition by generator substitution

For a general monomial-leading $G = cX^\rho(1 + h)$, Taylor expansion about X is not appropriate when $\rho \neq 1$. Use the generator algorithm:

Algorithm 6.7 (Generator composition). *To compute $F \circ G$ below t^N :*

1. *Compute $v = (1 + h)^{-1}$ and $\lambda = \log(1 + h)$ to the required precision.*
2. *For each input block $P_n(\ell)t^n$ that can reach order below N , form the finite polynomial shift*

$$P_n(\log c + \rho\ell + \lambda) = \sum_{j=0}^{\deg P_n} \frac{P_n^{(j)}(\log c + \rho\ell)}{j!} \lambda^j.$$

3. *Multiply by $c^{-n}t^{\rho n}v^n$, truncate, and accumulate.*

Negative n require positive powers of $1 + h$, which are finite integer powers. Positive n require reciprocal powers, computed by binomial series or repeated multiplication of v .

6.8 Associativity and substitution order

If all compositions involved are admissible, then

$$(F \circ G) \circ H = F \circ (G \circ H). \quad (6.15)$$

In terms of substitution maps,

$$\Phi_H \circ \Phi_G = \Phi_{G \circ H}. \quad (6.16)$$

Notice the order: Φ_G means substitution of G on the right. Equation (6.16) is a useful test when implementing nested composition.

For near-linear maps

$$G = X + A, \quad H = X + B,$$

we have

$$G \circ H = X + B + A \circ (X + B). \quad (6.17)$$

The correction remains in $\mathcal{R}$, so near-linear maps are closed under composition.

6.9 Chain rule and Faa di Bruno formula

Formal differentiation commutes with admissible composition:

$$D_X(F \circ G) = (F' \circ G)G'. \quad (6.18)$$

Higher derivatives are governed by the partial Bell polynomials:

$$D_X^n(F \circ G) = \sum_{k=1}^n (F^{(k)} \circ G) B_{n,k}(G', G'', \dots, G^{(n-k+1)}). \quad (6.19)$$

In the near-identity setting, each additional derivative improves the t -valuation, so the combinatorial size of Equation (6.19) is often more important than any support issue.

6.10 Where the one-logarithm class ceases to be closed

The closure theorem has sharp and informative boundaries.

6.10.1 Fractional powers

If $G \sim X^{1/2}$, then $t \circ G = t^{1/2}$, so a Puiseux extension is required. The logarithmic coefficient remains harmless: $\ell \circ G = \ell/2 + o(1)$.

6.10.2 A logarithmic factor in the leading monomial

If

$$G \sim X^\rho \ell^\sigma,$$

then

$$\log G \sim \rho \ell + \sigma \log \ell. \quad (6.20)$$

Unless $\sigma = 0$, the new scale $\log \ell = \log \log X$ appears. The one-logarithm ring must be enlarged to polynomial coefficients in another iterated logarithm, or to a full logarithmic transseries class.

6.10.3 Exponential inner functions

If $G = e^X$, then $t \circ G = e^{-X}$, an exponentially small monomial. Power-log transseries do not contain it. This is a height-changing composition naturally handled by logarithmic-exponential transseries.

6.10.4 Composition into a small argument

An expansion of $F(Y)$ as $Y \rightarrow +\infty$ cannot be composed with an inner series $G(X) \rightarrow 0$ without analytic continuation or a separate expansion of F near zero. Formal symbols do not erase the asymptotic base point.

Closure is scale-dependent

An expression may be algebraically meaningful as a function while its formal composition leaves the selected transseries class. This is not a failure of transseries; it is information that the operation reveals a missing scale. Fractional powers call for Puiseux supports, leading log factors call for iterated logs, and exponentials call for higher transseries height.

6.11 The near-identity substitution monoid

Define

$$\mathcal{G} := \{X + A : A \in \mathcal{R}\}. \quad (6.21)$$

Equation [Equation \(6.17\)](#) shows that $\mathcal{G}$ is a monoid under composition, with identity X . In [Chapter 7](#) we prove that every element has a unique inverse in the same class, so it is actually a group. The action

$$\mathcal{G} \curvearrowright \mathcal{P}, \quad (G, F) \mapsto F \circ G,$$

is by continuous algebra automorphisms.

Two equivalent composition engines

For $G = X + A$, composition can be computed either by transforming the generators

$$t \mapsto t/(1 + tA), \quad \ell \mapsto \ell + \log(1 + tA),$$

or by the Taylor series

$$F \circ (X + A) = \sum_{j \geq 0} A^j F^{(j)} / j!.$$

The generator method exposes scale changes; the Taylor method is often simpler for near-identity calculations and proves the contraction properties needed for reversion.

Part IV

Series reversal at infinity

Chapter 7

Compositional inversion of near-identity transseries

7.1 Reciprocal versus reverse

Two operations that are both casually called “inversion” must be separated. For a nonzero transseries F , the multiplicative reciprocal F^{-1} satisfies

$$FF^{-1} = 1.$$

The compositional inverse, or *reverse*, $F^{[-1]}$, satisfies

$$F \circ F^{[-1]} = F^{[-1]} \circ F = X. \quad (7.1)$$

Division was treated in [Chapter 5](#). This chapter treats reversal of maps asymptotic to the identity:

$$F = X + \phi, \quad \phi \in \mathcal{R}. \quad (7.2)$$

Because $\phi = o(X)$, such a map is eventually increasing in ordinary real situations: $F' = 1 + \phi' = 1 + o(1)$. Formally, $F' \in 1 + t\mathcal{R}$ is a unit.

7.2 The inverse equation as a fixed point

Write the unknown inverse as

$$H = X + \psi, \quad \psi \in \mathcal{R}.$$

The equation $F \circ H = X$ becomes

$$\psi = -\phi \circ (X + \psi). \quad (7.3)$$

Define

$$\mathcal{T}(\psi) := -\phi \circ (X + \psi). \quad (7.4)$$

The crucial observation is that $\mathcal{T}$ improves the order of a difference.

Lemma 7.1 (One-order contraction). *For $U, V \in \mathcal{R}$,*

$$\nu_t(\mathcal{T}(U) - \mathcal{T}(V)) \geq \nu_t(U - V) + 1. \quad (7.5)$$

Proof. Use Taylor composition around $X + V$:

$$\phi \circ (X + U) - \phi \circ (X + V) = \sum_{j \geq 1} \frac{(U - V)^j}{j!} \phi^{(j)} \circ (X + V).$$

The $j = 1$ term has valuation at least $v_t(U - V) + 1$, because $\phi' \in t\mathcal{R}$. For $j \geq 2$, the valuation is at least $jv_t(U - V) + j$, and in particular no smaller when $v_t(U - V) \geq 0$. Negating the difference does not change its valuation. $\square$

Theorem 7.2 (Near-identity reversion theorem). *Every $F = X + \phi$ with $\phi \in \mathcal{R}$ has a unique compositional inverse*

$$F^{[-1]} = X + \psi, \quad \psi \in \mathcal{R}.$$

It is the t -adic limit of the Picard iterates

$$\psi^{[0]} = 0, \quad \psi^{[r+1]} = -\phi \circ (X + \psi^{[r]}). \quad (7.6)$$

Moreover,

$$v_t(\psi^{[r+1]} - \psi^{[r]}) \geq r. \quad (7.7)$$

Proof. The first difference $\psi^{[1]} - \psi^{[0]} = -\phi$ has nonnegative valuation. Repeated application of [Theorem 7.1](#) proves [Equation \(7.7\)](#), so the sequence is Cauchy and converges in the complete ring $\mathcal{R}$. Continuity of composition gives the fixed-point equation, hence $F \circ (X + \psi) = X$.

If U and V are two fixed points, then

$$v_t(U - V) = v_t(\mathcal{T}(U) - \mathcal{T}(V)) \geq v_t(U - V) + 1,$$

which is impossible for a nonzero difference. Thus the right inverse is unique. Applying the same construction to it gives a right inverse of H ; associativity then shows that the unique right inverse is also a left inverse. $\square$

Corollary 7.3. *The set*

$$\mathcal{G} = \{X + A : A \in \mathcal{R}\}$$

is a group under composition.

7.3 Coefficient-by-coefficient reversal

The contraction theorem contains a triangular coefficient algorithm. Write

$$\psi = \sum_{n \geq 0} \psi_n(\ell) t^n. \quad (7.8)$$

Suppose $\psi_0, \dots, \psi_{N-1}$ are known and define

$$\psi_{<N} = \sum_{n < N} \psi_n(\ell) t^n.$$

Then

$$\psi_N(\ell) = -[t^N] \phi \circ (X + \psi_{<N}). \quad (7.9)$$

The unknown $\psi_N t^N$ cannot affect the right side at order t^N : it enters the transformed logarithm through $t\psi$, one order later, and every sensitivity of ϕ carries a derivative factor of order t .

At order zero, the rule is especially simple:

$$\psi_0(\ell) = -\phi_0(\ell). \quad (7.10)$$

Algorithm 7.4 (Triangular series reversal). *Given $F = X + \phi$ and a target order N :*

1. *Set $\psi_{<0} = 0$.*
2. *For $n = 0, 1, \dots, N-1$, compute*

$$c_n = [t^n] \phi \circ (X + \psi_{<n})$$

and append $-c_n t^n$ to $\psi_{<n}$.

3. *Return $H = X + \psi_{<N}$.*

Then $F \circ H = X + O_t(t^N)$.

Picard iteration and triangular extraction are the same mechanism organized differently. Picard is concise and lazy; triangular extraction makes each new coefficient explicit.

7.4 Newton reversion

For high precision, solve

$$\mathcal{F}(H) := F \circ H - X = 0$$

by Newton's method. Linearizing in a correction Δ gives

$$F \circ (H + \Delta) - X = \mathcal{F}(H) + (F' \circ H)\Delta + \text{higher order}.$$

Hence

$$H_{\text{new}} = H - \frac{F \circ H - X}{F' \circ H}. \quad (7.11)$$

The denominator lies in $1 + t\mathcal{R}$ and is invertible by [Theorem 5.1](#).

Theorem 7.5 (Quadratic improvement). *Suppose $H = X + \eta$ and*

$$F \circ H - X = O_t(t^m).$$

Then the Newton update [Equation \(7.11\)](#), computed consistently to sufficient precision, satisfies

$$F \circ H_{\text{new}} - X = O_t(t^{2m}).$$

In this particular scale the order is often even higher because $F'' \in t^2\mathcal{R}$.

Proof. Set $E = F \circ H - X$ and $\Delta = -E/(F' \circ H)$. Since the denominator is a unit, $v_t \Delta \geq m$. Taylor expansion gives

$$F \circ (H + \Delta) - X = E + (F' \circ H)\Delta + \sum_{j \geq 2} \frac{\Delta^j}{j!} (F^{(j)} \circ H).$$

The first two terms cancel. The remaining terms have valuation at least $jm + j \geq 2m$, since $F^{(j)} \in t^j\mathcal{R}$ for $j \geq 2$. $\square$

Algorithm 7.6 (Newton reversion to order N). *Start with any $H_0 = X + \eta_0$ correct modulo t , usually $\eta_0 = -\phi_0(\ell)$. At precisions $1, 2, 4, 8, \dots$, compute*

$$E = F \circ H - X, \quad J = F' \circ H, \quad H \leftarrow H - E/J,$$

truncating every operation to the new precision. Finish with a residual check $F \circ H - X = O_t(t^N)$.

7.5 Lagrange–Bürmann inversion at infinity

There is a closed universal formula for the inverse correction. Its form is classical, but the justification for setting its auxiliary parameter equal to one is particularly transparent in the present filtration.

Theorem 7.7 (At-infinity Lagrange–Bürmann formula). *Let $\phi \in \mathcal{R}$. Then*

$$(X + \phi)^{[-1]} = X + \sum_{m=1}^{\infty} \frac{(-1)^m}{m!} D_X^{m-1}(\phi^m). \quad (7.12)$$

The sum is t -adically well defined. Modulo t^N , only $m \leq N$ can contribute.

Proof. Introduce a bookkeeping parameter ε and let

$$F_\varepsilon(X) = X + \varepsilon\phi(X), \quad F_\varepsilon^{[-1]}(Y) = Y + u(Y, \varepsilon).$$

The inverse equation is

$$u = -\varepsilon\phi(Y + u). \quad (7.13)$$

Classical formal Lagrange inversion in the parameter ε yields

$$u(Y, \varepsilon) = \sum_{m \geq 1} \frac{(-\varepsilon)^m}{m!} D_Y^{m-1}(\phi(Y)^m). \quad (7.14)$$

Ordinarily, a formal power series in ε cannot simply be evaluated at $\varepsilon = 1$. Here it can, because

$$\nu_t(D_X^{m-1}(\phi^m)) \geq m - 1.$$

Thus the m -th term moves to arbitrarily high t -order, and [Theorem 2.3](#) applies. Setting $\varepsilon = 1$ gives [Equation \(7.12\)](#). $\square$

Remark 7.8. *The theorem is stronger than a mnemonic. It converts reversion into finitely many differentiations and multiplications at each requested order, with no implicit coefficient solving. It also exposes structural information, such as degree growth, that is less visible in fixed-point iteration.*

7.6 Finite coefficient formula

If

$$\phi = \sum_{r \geq 0} \phi_r(\ell) t^r,$$

then the coefficient of t^N in the inverse correction is

$$[t^N]((X + \phi)^{[-1]} - X) = \sum_{m=1}^{N+1} \frac{(-1)^m}{m!} [t^N] D_X^{m-1}(\phi^m). \quad (7.15)$$

Every term on the right is a finite polynomial calculation. Formula [Equation \(7.15\)](#) is convenient for generating one coefficient in isolation or proving identities.

7.7 Pure polynomial-logarithmic perturbations

A particularly important family is

$$F = X + P(\ell), \quad P \in \mathbb{K}[\ell]. \quad (7.16)$$

The $m = 1$ term in Equation (7.12) is $-P(\ell)$. Put

$$F^{[-1]} = X - P(\ell) + \sum_{n \geq 1} Q_n(\ell) t^n. \quad (7.17)$$

Taking $m = n + 1$ in Equation (7.12) and using Equation (3.4) gives the closed formula

$$Q_n(\ell) = \frac{(-1)^{n+1}}{(n+1)!} D(D-1) \cdots (D-n+1) P(\ell)^{n+1}, \quad D = D_\ell. \quad (7.18)$$

Theorem 7.9 (Degree law for polynomial perturbations). *Let $P(\ell) = a_d \ell^d + \text{lower powers}$, with $d \geq 1$ and $a_d \neq 0$. Then for $n \geq 1$,*

$$\deg Q_n = d(n+1) - 1, \quad [\ell^{d(n+1)-1}] Q_n = \frac{d}{n} a_d^{n+1}. \quad (7.19)$$

Proof. The polynomial P^{n+1} has degree $d(n+1)$ and leading coefficient a_d^{n+1} . The first factor D in Equation (7.18) lowers the degree by one and multiplies the leading coefficient by $d(n+1)$. Each later factor $D-j$, $j = 1, \dots, n-1$, preserves that degree through its $-j$ part and multiplies the leading coefficient by $-j$. The operator therefore has leading coefficient

$$d(n+1) a_d^{n+1} (-1)^{n-1} (n-1)!.$$

Multiplication by $(-1)^{n+1}/(n+1)!$ leaves da_d^{n+1}/n , with positive sign. $\square$

The rapid degree growth for $d > 1$ is a warning for symbolic computation: although only N coefficient blocks are required, their internal polynomial degrees can reach approximately dN .

7.8 Affine logarithmic perturbations

For

$$F = X + a\ell + b, \quad a \neq 0, \quad (7.20)$$

formula Equation (7.18) begins

$$\begin{aligned} F^{[-1]} &= X - a\ell - b + a(a\ell + b)t \\ &\quad + \frac{a(a\ell + b)(a\ell - 2a + b)}{2} t^2 \\ &\quad + \frac{a(a\ell + b)}{6} (2a^2\ell^2 - 9a^2\ell + 4ab\ell + 6a^2 - 9ab + 2b^2) t^3 + O_t(t^4). \end{aligned} \quad (7.21)$$

This family has an exact Lambert representation. Solving $Y = X + a \log X + b$ gives

$$X = aW\left(a^{-1} \exp\left(\frac{Y-b}{a}\right)\right), \quad (7.22)$$

with branches chosen consistently. Formula Equation (7.21) is the large- Y expansion of Equation (7.22).

7.9 Perturbations that already contain inverse powers

Nothing in [Theorem 7.7](#) requires ϕ to depend only on ℓ . For example, take

$$F = X + \ell + at. \quad (7.23)$$

The inverse has correction

$$\begin{aligned} F^{[-1]} - X &= -\ell + (\ell - a)t \\ &\quad + \left(\frac{\ell^2}{2} - \ell - a\ell + a \right) t^2 \\ &\quad + \left(\frac{\ell^3}{3} - \frac{3}{2}\ell^2 + \ell - a\ell^2 + 3a\ell - a - a^2 \right) t^3 + O_t(t^4). \end{aligned} \quad (7.24)$$

One may derive this by the Lagrange formula, triangular extraction, or Newton reversion. Substitution into F verifies that the residual is $O_t(t^4)$.

7.10 Normalization beyond leading coefficient one

Suppose

$$F(X) = cX + \phi(X), \quad c \in \mathbb{K}^\times, \quad \phi \in \mathcal{R}. \quad (7.25)$$

Let $G(U) = U + c^{-1}\phi(U)$. Then

$$F(X) = cG(X), \quad F^{[-1]}(Y) = G^{[-1]}(Y/c).$$

The final substitution $Y \mapsto Y/c$ shifts the logarithm by $-\log c$. Thus affine leading behavior is handled by scaling, followed by near-identity reversion.

If the leading behavior is cX^ρ with $\rho \neq 1$, the leading inverse contains $X^{1/\rho}$, and a Puiseux or Hahn extension is generally required. The reversion principle survives, but the convenient integer t -filtration must be replaced by one adapted to the fractional leading scale.

7.11 Left inverse, right inverse, and residuals

In a noncommutative composition setting, checking only one side can seem insufficient. The uniqueness theorem resolves the issue, but computationally it is cheap to check both

$$F \circ H - X = O_t(t^N), \quad (7.26)$$

$$H \circ F - X = O_t(t^N). \quad (7.27)$$

The two residuals catch different implementation errors, especially incorrect substitution order.

For analytic realizations with $F'(X) = 1 + o(1)$, a small composition residual also controls the inverse error. Roughly, if $F(H_N(X)) - X = O(X^{-N}(\log X)^M)$, the mean value theorem transfers the same order to $H_N - F^{-1}$, because F' stays bounded away from zero.

7.12 Comparison of the four methods

Method	Main strength	Typical use
Triangular coefficients	Transparent dependence on earlier blocks	Hand derivations and exact generation of one new coefficient at a time.
Picard fixed point	Minimal setup; one new block per iteration	Lazy streams and proofs by contraction.
Newton reversion	Quadratic precision growth	High-order computer algebra with fast multiplication and composition.
Lagrange–Bürmann	Closed universal formula	Structural identities, degree laws, and direct coefficient formulas.

No method is universally best. A robust implementation often uses a simple triangular or Lagrange method for low orders, switches to Newton at high precision, and verifies the result by independent composition.

Reversion principle

The derivative of every $\phi \in \mathcal{R}$ lies in $t\mathcal{R}$. Therefore a change in the guessed inverse is damped by one fast order when passed through ϕ . This single filtration fact yields existence, uniqueness, coefficient stabilization, Newton acceleration, and the formal summability of the Lagrange–Bürmann inverse series.

Chapter 8

The Wright omega inverse and its universal polynomials

8.1 The canonical logarithmic reverse

Consider the simplest nontrivial member of the near-identity group:

$$T(X) = X + \log X = X + \ell. \quad (8.1)$$

Its real inverse is the Wright omega function ω , characterized by

$$\omega(X) + \log \omega(X) = X. \quad (8.2)$$

On the positive real branch, $\omega(X) = W(e^X)$. Globally in the complex plane, branch and unwinding conventions must be supplied, but the formal expansion at a chosen large branch is local and unambiguous.

By [Theorem 7.2](#),

$$\omega(X) = X - \ell + \sum_{n \geq 1} P_n(\ell) t^n, \quad t = X^{-1}. \quad (8.3)$$

The polynomials P_n are universal: the same sequence appears in every branch of the large-argument Lambert expansion after the correct branch logarithm is inserted.

8.2 Derivation from Lagrange–Bürmann inversion

Set $\phi = \ell$ in [Equation \(7.12\)](#). The term with $m = 1$ is $-\ell$. For $n \geq 1$, the coefficient of t^n comes from $m = n + 1$:

$$P_n(u) = \frac{(-1)^{n+1}}{(n+1)!} D(D-1) \cdots (D-n+1) u^{n+1}, \quad D = \frac{d}{du}. \quad (8.4)$$

The box contains an arbitrary-order coefficient formula involving only a falling factorial of the differentiation operator.

The first cases illustrate the mechanism:

$$\begin{aligned} P_1(u) &= \frac{1}{2} D u^2 = u, \\ P_2(u) &= -\frac{1}{6} D(D-1) u^3 = \frac{u^2}{2} - u, \\ P_3(u) &= \frac{1}{24} D(D-1)(D-2) u^4 = \frac{u^3}{3} - \frac{3}{2} u^2 + u. \end{aligned}$$

Substitution into Equation (8.3) reproduces the beginning of Equation (1.3) after $X = \log z$.

8.3 Stirling cycle numbers

The signed Stirling numbers of the first kind $s(n, k)$ are defined by

$$x(x-1)\cdots(x-n+1) = \sum_{k=0}^n s(n, k)x^k. \quad (8.5)$$

The unsigned Stirling cycle numbers are

$$\left[\begin{matrix} n \\ k \end{matrix} \right] = (-1)^{n-k} s(n, k). \quad (8.6)$$

Replacing x by the differential operator D in Equation (8.5) and applying it to u^{n+1} turns Equation (8.4) into an explicit coefficient sum.

Theorem 8.1 (Stirling formula for the reversion polynomials). *For $n \geq 1$,*

$$P_n(u) = \sum_{m=1}^n (-1)^{n-m} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] \frac{u^m}{m!}. \quad (8.7)$$

Proof. Expand the operator in Equation (8.4):

$$D(D-1)\cdots(D-n+1) = \sum_{k=0}^n s(n, k)D^k.$$

The $k=0$ term vanishes because $s(n, 0) = 0$ for $n \geq 1$, and

$$D^k u^{n+1} = \frac{(n+1)!}{(n+1-k)!} u^{n+1-k}.$$

Set $m = n+1-k$. Then

$$\begin{aligned} P_n(u) &= (-1)^{n+1} \sum_{m=1}^n s(n, n-m+1) \frac{u^m}{m!} \\ &= \sum_{m=1}^n (-1)^{n-m} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] \frac{u^m}{m!}, \end{aligned}$$

where the signs differ by an even exponent. □

8.4 Immediate structural consequences

Formula Equation (8.7) gives several facts at once.

Proposition 8.2 (Basic properties). *For $n \geq 1$:*

1. $P_n(0) = 0$ and $P'_n(0) = (-1)^{n-1}$.
2. $\deg P_n = n$, with leading coefficient $1/n$.

3. The nonzero coefficients alternate in sign: $[u^m]P_n$ has sign $(-1)^{n-m}$.
4. The coefficient of u^{n-1} is $-H_{n-1}$, where $H_r = 1 + 1/2 + \cdots + 1/r$.
5. The coefficient of u^2 is $(-1)^{n-2}n(n-1)/4$.
6. $n!P_n(u) \in \mathbb{Z}[u]$.

Proof. The constant term is absent. For $m = 1$, use $\begin{bmatrix} n \\ n \end{bmatrix} = 1$. For $m = n$, use $\begin{bmatrix} n \\ 1 \end{bmatrix} = (n-1)!$, giving $(n-1)!/n! = 1/n$. Positivity of unsigned Stirling numbers proves the sign assertion. For $m = n-1$, use $\begin{bmatrix} n \\ 2 \end{bmatrix} = (n-1)!H_{n-1}$. For $m = 2$, use $\begin{bmatrix} n \\ n-1 \end{bmatrix} = \binom{n}{2}$. Finally, $m! \mid n!$ for $m \leq n$, so multiplying Equation (8.7) by $n!$ clears all denominators. $\square$

The leading coefficient $1/n$ also follows from the general degree theorem [Theorem 7.9](#) with $P(u) = u$.

8.5 A differential-integral recurrence

The polynomials can be generated without Stirling tables.

Theorem 8.3 (Recurrence). *Starting with $P_1(u) = u$,*

$$P'_{n+1}(u) = nP_n(u) - P'_n(u), \quad (8.8)$$

$$P_{n+1}(0) = 0. \quad (8.9)$$

Equivalently,

$$P_{n+1}(u) = -P_n(u) + n \int_0^u P_n(v) dv. \quad (8.10)$$

Proof. Let

$$U(t, u) = \sum_{n \geq 1} P_n(u) t^n. \quad (8.11)$$

We derive in the next section that U satisfies

$$U = -\log(1 - t(u - U)).$$

Implicit differentiation with respect to u and t , followed by elimination of the common denominator, gives the first-order partial differential equation

$$(1 + t)U_u = t + t^2 U_t. \quad (8.12)$$

Equating the coefficient of t^{n+1} yields $P'_{n+1} = nP_n - P'_n$. Since every P_n has zero constant term by [Equation \(8.4\)](#), integration from zero gives [Equation \(8.10\)](#). $\square$

The recurrence is computationally attractive: one integration and one subtraction produce the next polynomial, and all coefficients remain rational.

8.6 The implicit ordinary generating equation

Write the omega expansion as

$$\omega(X) = X - u + U, \quad u = \log X, \quad U = \sum_{n \geq 1} P_n(u) X^{-n}. \quad (8.13)$$

Then

$$\omega(X) = X(1 - X^{-1}(u - U)),$$

so

$$\log \omega(X) = u + \log(1 - X^{-1}(u - U)).$$

Substitution into $\omega + \log \omega = X$ yields

$$\boxed{U = -\log(1 - t(u - U))}. \quad (8.14)$$

This one equation recursively generates every P_n . Expanding the right side,

$$U = \sum_{k \geq 1} \frac{t^k (u - U)^k}{k}. \quad (8.15)$$

At order t^n , only $P_1, \dots, P_{n-1}$ occur on the right, giving a triangular recurrence.

Example 8.4 (Extracting the first three polynomials). *From Equation (8.15):*

$$\begin{aligned} [t]U &= u, \\ [t^2]U &= -P_1 + \frac{u^2}{2}, \\ [t^3]U &= -P_2 + \frac{P_1^2}{2} - uP_1 + \frac{u^3}{3}. \end{aligned}$$

Using $P_1 = u$, this gives

$$P_2 = \frac{u^2}{2} - u, \quad P_3 = \frac{u^3}{3} - \frac{3}{2}u^2 + u.$$

8.7 First eight polynomials

For reference,

$$\begin{aligned} P_1(u) &= u, \\ P_2(u) &= \frac{u^2}{2} - u, \\ P_3(u) &= \frac{u^3}{3} - \frac{3}{2}u^2 + u, \\ P_4(u) &= \frac{u^4}{4} - \frac{11}{6}u^3 + 3u^2 - u, \\ P_5(u) &= \frac{u^5}{5} - \frac{25}{12}u^4 + \frac{35}{6}u^3 - 5u^2 + u, \\ P_6(u) &= \frac{u^6}{6} - \frac{137}{60}u^5 + \frac{75}{8}u^4 - \frac{85}{6}u^3 + \frac{15}{2}u^2 - u, \\ P_7(u) &= \frac{u^7}{7} - \frac{49}{20}u^6 + \frac{203}{15}u^5 - \frac{245}{8}u^4 + \frac{175}{6}u^3 - \frac{21}{2}u^2 + u, \\ P_8(u) &= \frac{u^8}{8} - \frac{363}{140}u^7 + \frac{3283}{180}u^6 - \frac{6769}{120}u^5 + \frac{245}{3}u^4 - \frac{161}{3}u^3 + 14u^2 - u. \end{aligned} \quad (8.16)$$

Additional polynomials appear in [Chapter B](#).

8.8 Fixed-point iteration and stabilization

Equation (8.14) defines the map

$$\mathcal{S}(U) = -\log(1 - t(u - U)).$$

Its formal derivative with respect to U is

$$\frac{\partial \mathcal{S}}{\partial U} = -\frac{t}{1 - t(u - U)} \in t\mathcal{R}. \quad (8.17)$$

Therefore one iteration gains one power of t . Starting with $U_0 = 0$,

$$U_{r+1} = -\log(1 - t(u - U_r)), \quad (8.18)$$

produces the first r stable coefficient blocks after r iterations. This is the Lambert-specific version of Theorem 7.2.

8.9 Residual equation

For a truncation

$$\omega_N = X - u + \sum_{n=1}^N P_n(u) X^{-n},$$

define

$$R_N = \omega_N + \log \omega_N - X. \quad (8.19)$$

The coefficient construction is equivalent to

$$R_N = \mathcal{O}_t(t^{N+1}). \quad (8.20)$$

This residual is often easier to compute than a direct comparison with an exact special-function value. One Newton correction for $y + \log y - X = 0$ is

$$\omega_N^{\text{new}} = \omega_N - \frac{\omega_N + \log \omega_N - X}{1 + 1/\omega_N}. \quad (8.21)$$

It approximately doubles the number of correct asymptotic blocks and is also an excellent numerical polishing step.

8.10 Three independent derivations

The same polynomial sequence has now appeared from:

1. the general Lagrange–Bürmann reverse of $X + \log X$, yielding the operator formula Equation (8.4);
2. expansion of the fixed-point equation Equation (8.14);
3. the differential-integral recurrence Equation (8.10).

Agreement among these routes is more than aesthetic. It provides independent implementation strategies and makes transcription errors in high-order coefficients easy to detect.

Universal Lambert reversion polynomials

The inverse of $X + \log X$ is controlled by one polynomial sequence P_n . The operator formula explains its differential origin, the Stirling formula gives explicit coefficients, the integral recurrence generates it in linear stages, and the implicit generating equation connects it directly to the inverse problem.

Chapter 9

Lambert W from the omega reverse

9.1 Reduction to the canonical inverse

Return to

$$W(z)e^{W(z)} = z.$$

On a branch where logarithms are chosen consistently, taking a logarithm gives

$$W + \log W = \xi, \tag{9.1}$$

where ξ is the appropriate branch logarithm of z . Thus Lambert W is obtained by feeding a logarithm into the Wright omega inverse:

$$W(z) = \omega(\xi). \tag{9.2}$$

For the principal positive real branch, $\xi = \log z$. For the standard complex branch W_k , set

$$\xi_k = \text{Log } z + 2\pi i k, \quad k \in \mathbb{Z}, \tag{9.3}$$

with the next logarithm continued consistently in the sector under consideration.

Substitution of $X = \xi_k$, $u = \log \xi_k$ into [Equation \(8.3\)](#) gives the universal branch expansion.

Theorem 9.1 (Large-argument Lambert expansion). *As $|z| \rightarrow \infty$ in a branch-compatible sector,*

$$W_k(z) \sim \xi_k - \log \xi_k + \sum_{n=1}^{\infty} \frac{P_n(\log \xi_k)}{\xi_k^n}, \quad \xi_k = \text{Log } z + 2\pi i k. \tag{9.4}$$

Equivalently,

$$W_k(z) \sim \xi_k - \log \xi_k + \sum_{n=1}^{\infty} \frac{(-1)^n}{\xi_k^n} \sum_{m=1}^n \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] \frac{(-\log \xi_k)^m}{m!}. \tag{9.5}$$

Proof. Equation [Equation \(9.1\)](#) is exactly the defining equation $\omega + \log \omega = \xi_k$. Apply the omega expansion [Equation \(8.3\)](#). Substituting the Stirling formula [Equation \(8.7\)](#) and rearranging signs gives [Equation \(9.5\)](#). $\square$

Formula [Equation \(9.5\)](#) is the standard Stirling-cycle form recorded in the NIST Digital Library of Mathematical Functions and in the classical Lambert W literature [2, 5, 3].

9.2 The principal real expansion

For $z > 1$, put

$$L_1 = \log z, \quad L_2 = \log L_1. \quad (9.6)$$

Then

$$\begin{aligned} W_0(z) = & L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2^2/2 - L_2}{L_1^2} \\ & + \frac{L_2^3/3 - 3L_2^2/2 + L_2}{L_1^3} \\ & + \frac{L_2^4/4 - 11L_2^3/6 + 3L_2^2 - L_2}{L_1^4} \\ & + \frac{L_2^5/5 - 25L_2^4/12 + 35L_2^3/6 - 5L_2^2 + L_2}{L_1^5} + \dots \end{aligned} \quad (9.7)$$

The hierarchy is

$$1 \ll L_2 \ll L_1, \quad L_1^{-1} L_2^a = o(L_2^{-b}) \quad (a, b \text{ fixed}).$$

This is why all powers of L_2 at one inverse L_1 -order are grouped into a single coefficient polynomial.

9.3 A direct fixed-point derivation

Although the omega reduction is conceptually clean, it is useful to see the Lambert equation generate the same recursion directly. Write

$$W = L_1 - L_2 + U, \quad U = \sum_{n \geq 1} P_n(L_2) L_1^{-n}. \quad (9.8)$$

Since

$$W = L_1 \left(1 - \frac{L_2 - U}{L_1} \right),$$

we have

$$\log W = L_2 + \log \left(1 - \frac{L_2 - U}{L_1} \right).$$

Substitution into $W + \log W = L_1$ yields

$$U = -\log \left(1 - \frac{L_2 - U}{L_1} \right). \quad (9.9)$$

This is exactly Equation (8.14) with $t = L_1^{-1}$ and $u = L_2$. The formal derivative of the right side with respect to U is

$$-\frac{L_1^{-1}}{1 - L_1^{-1}(L_2 - U)},$$

which is of order L_1^{-1} . Each fixed-point iteration therefore locks in one additional inverse-logarithmic block.

9.4 Formal, asymptotic, and convergent status

The derivation above proves a formal identity in $\mathbb{K}[L_2][[L_1^{-1}]]$. Classical analysis adds more. The expansion is a Poincaré asymptotic expansion in standard large-argument sectors. Moreover, for sufficiently large $|z|$, the series in Equation (9.5) is absolutely convergent to $W_k(z)$; for the principal real branch it converges for $z \geq e$ [5, §4.13]. This convergence is a special strength of the Lambert series and should not be assumed for arbitrary transseries.

At $z = e$,

$$L_1 = 1, \quad L_2 = 0, \quad W_0(e) = 1.$$

Because $P_n(0) = 0$ for every n , the entire correction series vanishes and the expansion gives the exact value at the endpoint.

9.5 Numerical behavior

Let

$$W_N(z) = L_1 - L_2 + \sum_{n=1}^N \frac{P_n(L_2)}{L_1^n}, \quad (9.10)$$

with $W_0 = L_1 - L_2$. The following absolute errors were computed at high precision against the defining principal branch.

N	$z = 10$	$z = 10^6$	$z = e^{100}$
0	2.770×10^{-1}	1.936×10^{-1}	4.666×10^{-2}
1	8.524×10^{-2}	3.578×10^{-3}	6.051×10^{-4}
2	6.468×10^{-3}	7.263×10^{-4}	5.267×10^{-6}
3	7.778×10^{-3}	8.853×10^{-5}	8.154×10^{-8}
5	3.642×10^{-4}	1.247×10^{-6}	1.542×10^{-10}
8	1.115×10^{-4}	2.773×10^{-9}	2.938×10^{-15}
12	5.756×10^{-6}	4.990×10^{-14}	2.135×10^{-21}

The error need not decrease at every single truncation, as the $z = 10$ entries at $N = 2, 3$ show, even though the full real series converges. Farther into the asymptotic region, each additional block is dramatically effective.

9.6 Residual-based truncation

For numerical work, the best local quality indicator is often the defining equation rather than the last retained term. Given an approximation w , form

$$r(w) = w + \log w - L_1 \quad (9.11)$$

for the logarithmic equation, or

$$\tilde{r}(w) = we^w - z \quad (9.12)$$

when overflow and cancellation are not concerns. A Newton correction for the logarithmic equation is

$$w_{\text{new}} = w - \frac{w + \log w - L_1}{1 + 1/w}. \quad (9.13)$$

Starting from only a few transseries terms, one or two corrections usually produce full working precision. The logarithmic residual is preferable for astronomically large z , because it never forms e^w .

9.7 Derivative and arithmetic closure

Differentiating $We^W = z$ gives

$$W'(z) = \frac{W(z)}{z(1+W(z))}. \quad (9.14)$$

Once W is represented as a polynomial-logarithmic transseries in L_1^{-1} , the derivative follows from the differential rule, and the rational factor $W/(1+W)$ follows from division. For example,

$$W'(z) = \frac{1}{z} \left(1 - \frac{1}{L_1} + \frac{1-L_2}{L_1^2} + \frac{-L_2^2+3L_2-1}{L_1^3} + \dots \right). \quad (9.15)$$

This illustrates how the arithmetic chapters combine after one transseries is known.

9.8 The real branch W_{-1} near zero

For $-e^{-1} < x < 0$, there are two real branches. The lower branch satisfies $W_{-1}(x) \rightarrow -\infty$ as $x \rightarrow 0^-$. Define

$$\eta = \log(-1/x), \quad L = \log \eta. \quad (9.16)$$

Writing $x = -e^{-\eta}$ and $W_{-1}(x) = -v$ gives

$$ve^{-v} = e^{-\eta}, \quad v - \log v = \eta.$$

The same universal polynomials occur, with alternating inverse powers:

$$W_{-1}(x) \sim -\eta - L + \sum_{n=1}^{\infty} (-1)^n \frac{P_n(L)}{\eta^n} \quad (x \rightarrow 0^-). \quad (9.17)$$

The first terms are

$$\begin{aligned} W_{-1}(x) \sim & -\eta - L - \frac{L}{\eta} + \frac{L^2/2 - L}{\eta^2} \\ & - \frac{L^3/3 - 3L^2/2 + L}{\eta^3} + \frac{L^4/4 - 11L^3/6 + 3L^2 - L}{\eta^4} + \dots \end{aligned} \quad (9.18)$$

For orientation, the absolute errors of truncations of Equation (9.17) are:

N	$x = -10^{-3}$	$x = -10^{-6}$	$x = -10^{-20}$
0	2.776×10^{-1}	1.852×10^{-1}	8.152×10^{-2}
1	2.173×10^{-3}	4.855×10^{-3}	1.645×10^{-3}
2	3.537×10^{-3}	5.502×10^{-4}	7.550×10^{-6}
3	2.976×10^{-4}	8.753×10^{-5}	1.889×10^{-6}
5	1.810×10^{-5}	1.010×10^{-6}	3.038×10^{-9}
8	1.802×10^{-8}	2.427×10^{-9}	6.282×10^{-13}
12	3.809×10^{-10}	2.596×10^{-13}	3.377×10^{-18}

Again, a one-step increase need not improve the error monotonically at moderate arguments, but the hierarchy becomes rapidly effective as $x \rightarrow 0^-$.

9.9 Why the branch point needs a different transseries

The inverse-logarithmic expansion is adapted to $W \rightarrow \pm\infty$. Near the branch point $z = -e^{-1}$, one instead has $W \rightarrow -1$, and the derivative

$$W'(z) = \frac{W}{z(1+W)}$$

blows up because $1 + W = 0$. Set

$$p = \sqrt{2(1+ez)}. \quad (9.19)$$

Then the two local branches have Puiseux expansions

$$W_{0,-1}(z) = -1 \pm p - \frac{p^2}{3} \pm \frac{11p^3}{72} - \frac{43p^4}{540} + \dots \quad (9.20)$$

The square root is forced by the quadratic critical point of $w \mapsto we^w$. This is a canonical example of scale selection: polynomial-logarithmic transseries are correct at infinity and near the logarithmic singular limit $x \rightarrow 0^-$ on W_{-1} , while a Puiseux series is correct near $-e^{-1}$.

9.10 A practical high-precision recipe

For a large complex z on branch k :

1. Compute $\xi_k = \text{Log } z + 2\pi i k$ and a consistent $u = \log \xi_k$.
2. Generate $P_1, \dots, P_N$ by $P_{n+1} = -P_n + n \int_0^u P_n$, or use the Stirling formula.
3. Sum

$$w_N = \xi_k - u + \sum_{n=1}^N P_n(u) \xi_k^{-n}$$

with guarded precision.

4. Evaluate the logarithmic residual $r = w_N + \log w_N - \xi_k$, using the same branch of $\log$.
5. If numerical accuracy rather than a symbolic expansion is the goal, apply Newton's update [Equation \(9.13\)](#), or a higher-order Householder step.

The transseries provides a branch-aware starting value whose quality improves with $|\xi_k|$; Newton then removes the remaining local error.

Branch consistency

The algebraic polynomial sequence does not encode branch cuts. The quantities ξ_k , $\log \xi_k$, and the logarithm used in the residual must be continued consistently. An apparent failure of a high-order expansion is often a branch mismatch rather than a coefficient error.

Lambert structure in one line

Lambert W at large argument is the reverse of $X + \log X$ evaluated at a branch logarithm of the original argument:

$$W_k(z) = \omega(\xi_k), \quad \xi_k = \text{Log } z + 2\pi i k.$$

All coefficient polynomials, recurrences, and composition laws are therefore consequences of near-identity reversal in $\mathbb{K}[\log X](X^{-1})$.

Chapter 10

Beyond Lambert: reusable inverse templates

10.1 The general near-identity template

Suppose an inverse problem has been normalized to

$$Y = X + \phi(X), \quad \phi \in \mathcal{R}. \quad (10.1)$$

Then every method of [Chapter 7](#) applies without change. The most direct arbitrary-order formula is

$$X = Y + \sum_{m \geq 1} \frac{(-1)^m}{m!} D_Y^{m-1}(\phi(Y)^m). \quad (10.2)$$

Expanding the first four universal terms gives

$$\begin{aligned} X &= Y - \phi + \frac{1}{2}D(\phi^2) - \frac{1}{6}D^2(\phi^3) + \frac{1}{24}D^3(\phi^4) - \dots \\ &= Y - \phi + \phi\phi' - \left(\phi(\phi')^2 + \frac{1}{2}\phi^2\phi'' \right) \\ &\quad + \left(\phi(\phi')^3 + \frac{3}{2}\phi^2\phi'\phi'' + \frac{1}{6}\phi^3\phi''' \right) - \dots, \end{aligned} \quad (10.3)$$

where ϕ and its derivatives are evaluated at Y . The first displayed line is usually preferable: it preserves the compact differential structure and avoids the rapid growth of Bell-polynomial terms.

10.2 Polynomial perturbations in the logarithm

For

$$Y = X + P(\log X), \quad P \in \mathbb{K}[u],$$

the inverse is given by [Equations \(7.17\) and \(7.18\)](#). This family already covers nonlinear logarithmic corrections such as

$$Y = X + a(\log X)^2 + b \log X + c. \quad (10.4)$$

If $a \neq 0$, the coefficient of Y^{-n} in the inverse is a polynomial of degree $2n + 1$ in $\log Y$, with leading coefficient $2a^{n+1}/n$. The degree law predicts expression growth before any coefficient is calculated.

For example, with $P(u) = u^2$,

$$(X + \ell^2)^{[-1]} = X - \ell^2 + 2\ell^3 t + (\ell^5 - 5\ell^4) t^2 + \left(\frac{2}{3}\ell^7 - 7\ell^6 + 14\ell^5\right) t^3 + O_t(t^4). \quad (10.5)$$

The leading coefficients 2, 1, 2/3 agree with $2/n$ for $n = 1, 2, 3$, as predicted by [Theorem 7.9](#).

10.3 Adding lower inverse-power corrections

Consider

$$Y = X + P_0(\ell) + P_1(\ell)t + P_2(\ell)t^2 + \cdots. \quad (10.6)$$

The leading inverse correction is still $-P_0(\ell)$. Once it is removed, all lower terms are generated automatically by any of the four reversion methods. A useful organizational principle is:

1. treat each coefficient polynomial exactly;
2. use t -order, not log degree, as the outer precision;
3. truncate after every derivative, multiplication, and composition;
4. verify by composing the result back into the original map.

The inverse-power corrections interact with the logarithmic shift. For example, a term $P_1(\ell)/X$ is not reversed merely by changing its sign, because replacing X by $Y - P_0(\log Y) + \cdots$ changes both $1/X$ and $\log X$.

10.4 Conjugating an inverse problem into near-identity form

Not every natural map is asymptotic to X , but many can be transformed into one. Suppose

$$F = H^{-1} \circ (\text{id} + \phi) \circ H, \quad (10.7)$$

where H and H^{-1} are known on the relevant transseries class. Then

$$F^{[-1]} = H^{-1} \circ (\text{id} + \phi)^{[-1]} \circ H. \quad (10.8)$$

Taking logarithms or exponentials is a common instance. Edgar's general transseries inversion procedure uses iterated logarithms and exponentials to reduce a large positive transseries to a near-identity, log-free problem [7, 6].

10.5 The map $X \log X$

The equation

$$Y = X \log X \quad (10.9)$$

looks different from a near-identity perturbation. Put $u = \log X$, so $X = e^u$ and

$$Y = ue^u.$$

Therefore

$$u = W(Y), \quad X = e^{W(Y)} = \frac{Y}{W(Y)}. \quad (10.10)$$

Its leading inverse scale is

$$X \sim \frac{Y}{\log Y}, \quad (10.11)$$

not Y . Substituting the Lambert expansion into $Y/W(Y)$ and using the division rules gives

$$X = \frac{Y}{L_1} \left(1 + \frac{L_2}{L_1} + \frac{L_2^2 - L_2}{L_1^2} + \frac{L_2^3 - \frac{5}{2}L_2^2 + L_2}{L_1^3} + \cdots \right), \quad (10.12)$$

where $L_1 = \log Y$, $L_2 = \log \log Y$. This example shows how a non-near-identity inverse can still be built from polynomial-logarithmic arithmetic after the correct dominant inverse is found.

10.6 Power-log leading behavior

A broader model is

$$Y = cX^\alpha (\log X)^\beta, \quad c > 0, \quad \alpha > 0. \quad (10.13)$$

Taking logarithms and writing $u = \log X$ gives

$$\alpha u + \beta \log u = \log(Y/c). \quad (10.14)$$

When $\beta \neq 0$, this equation has an exact Lambert representation:

$$u = \frac{\beta}{\alpha} W\left(\frac{\alpha}{\beta} \exp\left(\frac{\log(Y/c)}{\beta}\right)\right), \quad (10.15)$$

with suitable branch choices. Dominant balance yields

$$X \sim c^{-1/\alpha} \alpha^{\beta/\alpha} Y^{1/\alpha} (\log Y)^{-\beta/\alpha}. \quad (10.16)$$

Corrections involve inverse powers of $\log Y$ with polynomials in $\log \log Y$. Fractional powers of Y and $\log Y$ show why the natural ambient object is now a generalized power-log or Hahn transseries, not the discrete near-identity ring alone.

10.7 Inverse problems with one more logarithmic level

Suppose the correction itself contains $\log \log X$, for example

$$Y = X + a \log X + b \log \log X. \quad (10.17)$$

Writing $t = X^{-1}$, $\ell_1 = \log X$, and $\ell_2 = \log \ell_1$, the same near-identity proof works in the coefficient ring

$$\mathbb{K}[\ell_1, \ell_2][[t]] \quad (10.18)$$

provided differentiation is defined by

$$D_X = t \partial_{\ell_1} + \frac{t}{\ell_1} \partial_{\ell_2} - t^2 \partial_t. \quad (10.19)$$

However, the coefficient $1/\ell_1$ is not polynomial in ℓ_1 , so a field such as $\mathbb{K}((\ell_1^{-1}))[\ell_2]((t))$ is more stable under differentiation. Each new iterated logarithm refines the coefficient field and its internal valuation.

10.8 Implicit equations and Newton polygons of scales

For a general implicit equation

$$\mathcal{E}(X, Y) = 0, \quad (10.20)$$

a practical transseries workflow is:

1. list the candidate dominant monomials of Y ;
2. choose a balance in which at least two largest terms of $\mathcal{E}$ cancel;
3. solve the resulting leading algebraic or exponential-logarithmic equation;
4. write $Y = Y_0 + U$ or $Y = Y_0(1 + U)$ with U small;
5. rewrite the equation as a contractive fixed point or apply formal Newton iteration;
6. enlarge the monomial scale whenever a required operation introduces a new power, logarithm, or exponential level.

This is the transseries analogue of Newton polygon reasoning. The difficult step is usually not coefficient extraction but choosing a scale closed under the transformed equation.

10.9 Analytic transfer to actual inverse functions

Formal reversion predicts the only possible asymptotic expansion of an actual inverse, but an analytic theorem is needed to prove realization. A typical sufficient setting is:

1. $F(X) = X + \phi(X)$ is real or complex analytic in an asymptotic domain;
2. ϕ admits the stated polynomial-logarithmic expansion uniformly in that domain;
3. $F'(X) = 1 + o(1)$, so F is locally invertible and its derivative is bounded away from zero;
4. composition and differentiation preserve the required remainder bounds.

Then a truncated formal inverse has a small composition residual, and an inverse-function estimate converts that residual into a remainder estimate for the true inverse. Symbolic algorithms for exp-log functions and their multiseries inverses develop this principle systematically [13, 12].

10.10 What remains universal and what changes

Across all these examples, three facts are stable:

1. composition is substitution on an ordered monomial hierarchy;
2. a fixed-point map is useful when it pushes errors to smaller monomials;
3. Newton iteration is useful when the formal derivative is a unit.

What changes is the ambient support: integer powers, fractional powers, iterated inverse logs, or exponentials. The polynomial-logarithmic ring is a particularly transparent local chart inside the larger transseries universe.

Inverse design rule

First find and factor out the dominant inverse scale. Only then ask for a near-identity correction. If the correction operations remain in $\mathbb{K}[\log X][[X^{-1}]]$, use the machinery of this article directly. If a fractional power, $\log \log X$, or e^{-X} appears, enlarge the monomial class rather than forcing the answer into an algebra that is not closed.

Part V

Algorithms, implementation, and extensions

Chapter 11

Symbolic implementation

11.1 Canonical data representation

A production implementation should not store a transseries as an unstructured expression tree. For a fixed truncation window, use a sparse ordered map

$$n \mapsto P_n(\ell), \quad (11.1)$$

where $n \in \mathbb{Z}$ is the t -exponent and P_n is a canonical univariate polynomial over a declared coefficient field. Zero polynomials are removed. The representation should carry:

1. a lower and upper t -cutoff;
2. the coefficient domain, such as $\mathbb{Q}[\ell]$, $\mathbb{Q}(\ell)$, or $\mathbb{Q}((\ell^{-1}))$;
3. the asymptotic variable and logarithmic-level metadata;
4. branch data when constants, logarithms, or fractional powers are complex.

Separating the outer support from the polynomial coefficient is important. The order comparison first inspects n , and only then the degree of P_n . Flattening everything into a bivariate polynomial loses this hierarchy and makes truncation ambiguous.

11.2 Exclusive precision convention

It is helpful to make precision an exclusive upper bound:

$$\text{truncate}_N(A) = \sum_{n < N} P_n(\ell) t^n. \quad (11.2)$$

Thus “precision N ” means equality modulo t^N . Every operation should specify how much input precision it consumes and should return a value tagged with the precision it actually guarantees.

For Laurent inputs with lower order r , multiplication modulo t^N requires the first factor through t^{N-s} when the second begins at t^s . Composition needs more careful backward precision planning because negative powers of the inner leading factor can amplify requested orders.

11.3 Core coefficient algorithms

With arrays indexed from zero, the basic recurrences are:

11.3.1 Multiplication

$$R_n = \sum_{j=0}^n P_j Q_{n-j}. \quad (11.3)$$

For sparse supports, iterate only over present pairs and discard sums outside the output window.

11.3.2 Division by a unit

If $B_0 = b \in \mathbb{K}^\times$,

$$C_n = b^{-1} \left(A_n - \sum_{j=1}^n B_j C_{n-j} \right). \quad (11.4)$$

At high precision, Newton inversion followed by one multiplication is usually preferable.

11.3.3 Near-identity composition

Either transform the generators

$$t_G = \frac{t}{1 + tA}, \quad \ell_G = \ell + \log(1 + tA), \quad (11.5)$$

and evaluate coefficient polynomials at ℓ_G , or use

$$F \circ (X + A) = \sum_{j=0}^{N-1} \frac{A^j}{j!} D_X^j F \pmod{t^N}. \quad (11.6)$$

The Taylor method has a simple stopping test when the derivative reaches the cutoff. The generator method generalizes more naturally to $cX^\rho(1+h)$.

11.3.4 Reversal

For low and moderate precision, use

$$\psi = \sum_{m=1}^N \frac{(-1)^m}{m!} D_X^{m-1}(\phi^m) \pmod{t^N}. \quad (11.7)$$

For high precision, switch to Newton reversion

$$H \leftarrow H - \frac{F \circ H - X}{F' \circ H} \quad (11.8)$$

with precision doubling.

11.4 Truncate early and often

Expression swell is the main practical enemy. Suppose a product is needed modulo t^N . Expanding an intermediate through t^{2N} and truncating at the end can multiply both runtime and memory by orders of magnitude, especially when coefficient degrees grow linearly with n . The safe discipline is:

1. truncate after every multiplication;

2. truncate after every derivative;
3. truncate powers while building them;
4. in a polynomial shift, stop powers of the small shift when their valuation reaches the cutoff;
5. normalize and remove zero coefficient blocks immediately.

For Newton algorithms, calculate each doubling step only to its new precision. Do not compute the final-precision products during early iterations.

11.5 A compact tested SymPy implementation

The following code is intentionally small rather than optimized. It represents a truncated element by a SymPy expression in t , L , implements both composition engines, computes a Newton reciprocal, performs Lagrange reversion, and verifies defining residuals. The code handles the nonnegative t -adic ring; a production Laurent implementation should use an explicit coefficient map.

```

1 from __future__ import annotations
2
3 from sympy import (
4     Symbol, diff, expand, factorial, integrate, log, series, simplify
5 )
6
7 t = Symbol("t")
8 L = Symbol("L")
9
10
11 def trunc(expr, order: int):
12     """Keep powers t**n with 0 <= n < order."""
13     if order <= 0:
14         return 0
15     return expand(series(expand(expr), t, 0, order).remove0())
16
17
18 def Dx(expr, order: int):
19     """d/dX when t=1/X and L=log(X), truncated in t."""
20     return trunc(t * diff(expr, L) - t**2 * diff(expr, t), order)
21
22
23 def reciprocal(B, order: int):
24     """Reciprocal when the t**0 coefficient is a nonzero scalar."""
25     b0 = expand(B).coeff(t, 0)
26     if b0 == 0 or L in b0.free_symbols:
27         raise ValueError("leading coefficient must be a nonzero scalar")
28
29     V = 1 / b0
30     precision = 1
31     while precision < order:
32         precision = min(2 * precision, order)
33         V = trunc(V * (2 - trunc(B, precision) * V), precision)
34     return trunc(V, order)
35
36
37 def compose_shift(F, A, order: int):

```

```

38     """Compute F(X+A) for F,A in K[L][[t]] by formal Taylor."""
39     result = 0
40     A_power = 1
41     derivative = trunc(F, order)
42
43     for j in range(order):
44         result = trunc(result + A_power * derivative / factorial(j), order)
45         A_power = trunc(A_power * A, order)
46         derivative = Dx(derivative, order)
47         if derivative == 0:
48             break
49
50     return result
51
52
53 def compose_generators(F, A, order: int):
54     """Same composition using transformed generators."""
55     t_image = trunc(t / (1 + t * A), order)
56     L_image = trunc(L + log(1 + t * A), order)
57     substituted = F.subs({t: t_image, L: L_image}, simultaneous=True)
58     return trunc(substituted, order)
59
60
61 def reverse_lagrange(phi, order: int):
62     """psi in (X+phi)^{-1} = X+psi modulo t^{**order}."""
63     psi = 0
64     for m in range(1, order + 1):
65         term = trunc(phi**m, order)
66         for _ in range(m - 1):
67             term = Dx(term, order)
68         psi = trunc(psi + (-1)**m * term / factorial(m), order)
69     return psi
70
71
72 def lambert_polynomials(count: int):
73     """Return [P_1(L), ..., P_count(L)]."""
74     if count < 1:
75         return []
76
77     result = [L]
78     for n in range(1, count):
79         Pn = result[-1]
80         Pnext = expand(-Pn + n * integrate(Pn, (L, 0, L)))
81         result.append(Pnext)
82     return result
83
84
85 def main() -> None:
86     order = 7
87     a = Symbol("a")
88
89     # Independent composition engines must agree.
90     F = L**2 + (L + 1)*t + (L**2 - L)*t**2 + (2*L + 1)*t**3
91     A = L

```

```

92  assert simplify(
93      compose_shift(F, A, 5) - compose_generators(F, A, 5)
94  ) == 0
95
96  # Reverse X + L + a/X and verify F(H)-X.
97  phi = L + a*t
98  psi = reverse_lagrange(phi, order)
99  residual = trunc(psi + compose_shift(phi, psi, order), order)
100 assert simplify(residual) == 0
101
102 # Verify a Newton reciprocal.
103 B = 1 + (L + 1)*t + L**2*t**2 + (L - 2)*t**3
104 V = reciprocal(B, order)
105 assert trunc(B*V - 1, order) == 0
106
107 for n, polynomial in enumerate(lambert_polynomials(8), start=1):
108     print(f"P_{n}(L) = {polynomial}")
109
110
111 if __name__ == "__main__":
112     main()

```

Listing 11.1: Reference implementation for the discrete polynomial-logarithmic ring.

11.6 Why the derivative implementation is correct

The line

```
return trunc(t * diff(expr, L) - t**2 * diff(expr, t), order)
```

is simply the chain rule

$$\frac{d}{dX} = \frac{dL}{dX} \partial_L + \frac{dt}{dX} \partial_t = t \partial_L - t^2 \partial_t.$$

Using this operator rather than differentiating after substituting $t = 1/X$, $L = \log X$ keeps expressions inside the canonical formal coordinates.

11.7 Independent tests

A strong test suite should check identities rather than hard-coded output alone. Useful properties include:

1. $\text{trunc}_N(A(B^{-1}) - 1) = 0$ for random unit series;
2. generator composition equals Taylor composition;
3. $(F \circ G) \circ H = F \circ (G \circ H)$ modulo the cutoff;
4. a computed reverse satisfies both $F \circ F^{[-1]} = X$ and $F^{[-1]} \circ F = X$;
5. Lambert polynomials generated by the integral recurrence agree with the Stirling formula;
6. the residual of the truncated omega expansion begins at the predicted order;
7. differentiation obeys product and chain rules modulo the cutoff.

Randomized property tests are particularly effective. Generate low-degree polynomial coefficients over $\mathbb{Q}$, use small cutoffs, and compare multiple algorithms. Exact rational arithmetic prevents numerical tolerance from hiding a wrong coefficient.

11.8 Complexity overview

Let $M(N)$ denote the cost of multiplying two truncated outer series, including coefficient-polynomial work. Then, schematically:

Operation	Typical cost	Comment
Naive multiplication	$O(N^2)$ coefficient products	Sparse support and degree bounds can reduce the constant.
Triangular reciprocal	$O(N^2)$	Simple and reliable at low order.
Newton reciprocal	$O(M(N))$ up to logarithmic factors	Requires precision-doubling multiplication.
Taylor composition	Up to $O(NM(N))$	Often less because derivatives reach the cutoff early.
Triangular/Picard reverse	Roughly one composition per new block	Excellent for lazy low-order output.
Newton reverse	A logarithmic number of compositions and divisions	Best when fast composition is available.
Lambert recurrence	$O(N^2)$ rational coefficient operations	Polynomial P_n has degree n .

These estimates suppress polynomial degree. For $X + P(\log X)$ with $\deg P = d$, the n -th inverse block has degree $d(n + 1) - 1$, so dense polynomial arithmetic can add another quadratic factor if implemented naively.

11.9 Lazy coefficient streams

The contraction proof supports a lazy interface. A request for coefficient t^N of a reverse needs only coefficients below N of the current approximation; higher coefficients are never inspected. One can therefore represent a transseries as a memoized stream of polynomial blocks and expose operations that request dependencies on demand.

Such an implementation should still carry an explicit support certificate. Without one, a generic recursive definition may ask for itself at the same or a larger monomial, causing nontermination. In the present near-identity class, the certificate is the one-order derivative gain $D_X \mathcal{R} \subseteq t\mathcal{R}$.

11.10 Branch and domain metadata

Real formal calculations often suppress branch information safely. Complex Lambert calculations cannot. A symbolic object representing $P_n(\log \xi_k)/\xi_k^n$ should record:

1. the branch index k of W_k ;

2. the branch used for $\text{Log } z$;
3. the continuation used for $\log \xi_k$;
4. the sector or path on which the asymptotic expansion is asserted.

Canonical algebra and analytic branch bookkeeping are separate layers. The polynomial coefficients are universal; the realization data select the function represented by the formal expression.

Implementation architecture

Use a sparse ordered map for outer powers, exact canonical polynomials for log coefficients, an exclusive cutoff, and a declared coefficient field. Make composition and reversion precision-aware, truncate every intermediate, and verify results by residual identities generated independently from the construction algorithm.

Chapter 12

Extensions of the polynomial-logarithmic class

12.1 The discrete ring as a local chart

The algebra $\mathbb{K}[\ell]((t))$ is not intended to contain every transseries. It is a particularly convenient chart for functions whose expansion at one large scale X uses integer inverse powers of X and finite nonnegative powers of $\log X$ in each block. Its advantages are substantial:

1. supports are indexed by ordinary integers;
2. every coefficient block is a finite polynomial;
3. the t -adic topology is complete and elementary;
4. differentiation raises outer order;
5. near-identity composition and reversal are closed.

Its limitations are equally informative. Each time an operation introduces a new monomial, the correct response is to enlarge the support group or the coefficient field in a controlled way.

12.2 Inverse powers of the logarithm

Division by a nonconstant polynomial in ℓ naturally leads to

$$\mathcal{F} = \mathbb{K}((\ell^{-1}))((t)). \quad (12.1)$$

An element has the form

$$\sum_{n \geq n_0} t^n \sum_{j \geq j_n} a_{n,j} \ell^{-j}, \quad (12.2)$$

where each coefficient of t^n may contain infinitely many decreasing powers of ℓ , but only finitely many positive powers. The asymptotic priority is lexicographic: first compare t -powers, then ℓ^{-1} -powers.

This field is appropriate for expressions such as

$$\frac{1}{\ell + 1 + t} = \frac{1}{\ell} - \frac{1}{\ell^2} + \cdots - t \left(\frac{1}{\ell^2} - \frac{2}{\ell^3} + \cdots \right) + \cdots .$$

Care is needed with nested truncation. One must specify both an outer t -cutoff and an inner ℓ^{-1} -cutoff, or use a monomial threshold in the full lexicographic order.

12.3 Puiseux-logarithmic series

To admit fractional powers of X , fix $q \geq 1$ and use

$$\mathbb{K}[\ell]((t^{1/q})). \quad (12.3)$$

The derivative formula generalizes without change:

$$D_X(t^\alpha P(\ell)) = t^{\alpha+1}(P' - \alpha P), \quad \alpha \in q^{-1}\mathbb{Z}. \quad (12.4)$$

A compositional inverse of a map with leading term cX^q naturally contains $X^{1/q}$, so Puiseux support is unavoidable.

Taking the union over all q admits rational exponents but loses a single discrete denominator bound. Hahn series provide a cleaner completion.

12.4 Hahn power-log series

Let $\Gamma \subseteq \mathbb{R}$ be an ordered additive group. A Hahn power-log series has schematic form

$$A = \sum_{\alpha \in S} P_\alpha(\ell) t^\alpha, \quad (12.5)$$

where $S \subseteq \Gamma$ is well ordered in the direction required for summation. Multiplication uses the Minkowski sum of supports. The well-order condition guarantees that each coefficient receives only an admissible family of contributions.

The differential rule is again

$$D_X(t^\alpha P) = t^{\alpha+1}(P' - \alpha P).$$

Near-identity composition remains controlled when the perturbation raises the chosen valuation. What was an integer “gain of one block” becomes a positive gain in Γ .

12.5 Power-log transseries near zero

Many dynamical applications are written near $x = 0^+$, using

$$x^\alpha (-\log x)^k. \quad (12.6)$$

Set $X = x^{-1}$. Then

$$x^\alpha (-\log x)^k = X^{-\alpha} (\log X)^k = t^\alpha \ell^k.$$

Thus the large- X framework is equivalent to a small- x power-log framework after inversion of the independent variable. Generalized Dulac series and the power-log transseries studied in local dynamics use supports of this kind, often with real exponents and integer log powers [10].

The direction of composition matters. A germ tangent to the identity near zero,

$$f(x) = x + o(x),$$

corresponds under $X = 1/x$ to a large-variable map with a specific near-identity correction. The formal group law can be transported between the two coordinates, but coefficient normalizations change.

12.6 Several iterated logarithms

Define

$$\ell_1 = \log X, \quad \ell_2 = \log \ell_1, \quad \dots, \quad \ell_r = \log \ell_{r-1}. \quad (12.7)$$

A finite-level logarithmic series may use coefficients polynomial or Laurent series in these variables. Their derivatives are

$$D_X \ell_j = \frac{1}{X \ell_1 \ell_2 \cdots \ell_{j-1}} = t (\ell_1 \ell_2 \cdots \ell_{j-1})^{-1}. \quad (12.8)$$

Therefore closure under differentiation already forces inverse powers of the earlier logarithms. A natural coefficient tower is constructed iteratively, for example

$$\mathbb{K}((\ell_r^{-1})) \cdots ((\ell_2^{-1}))((\ell_1^{-1}))((t)), \quad (12.9)$$

with an order matching

$$t \prec \ell_1^{-N} \prec \ell_2^{-N} \prec \cdots \prec 1.$$

The precise nesting convention matters; different iterated Laurent fields can encode different support orders.

Lambert W itself uses two logarithmic levels in the original argument z , but only one coefficient logarithm after the large variable is changed to $X = \log z$. This reduction is a useful general strategy.

12.7 Exponentially small and large monomials

Power-log series cannot represent e^{-X} , e^{-X^2} , or $e^{\sqrt{X}}$. Full logarithmic-exponential transseries adjoin monomials of the form

$$\exp(L), \quad (12.10)$$

where L is a purely large transseries, and then iterate logarithmic and exponential extensions. The resulting ordered field supports arithmetic, derivation, integration, and broad classes of composition [16, 6, 7].

In that larger setting, the polynomial-logarithmic ring appears as a low-height subalgebra. Its concrete formulas remain valid exactly; they are not replaced, only embedded into a richer support system.

12.8 Logarithmic-exponential Hardy fields

Actual germs generated from real constants and X by field operations, exponentials, and logarithms form classical logarithmic-exponential Hardy fields. Their growth comparison motivated early formal counterparts [9, 4, 14]. Transseries refine this growth structure by attaching canonical generalized series to many such germs.

The distinction is useful:

- a Hardy field consists of germs of actual functions;
- a transseries field consists of formal generalized series;
- an asymptotic expansion map relates the two when analytic realization is available.

Formal computations can therefore be exact in the transseries field even when questions of convergence, continuation, or summability remain analytic.

12.9 Grid-based and well-based support

Two common organizational frameworks for transseries support are *grid-based* and *well-based*. A grid-based support is generated by finitely many small monomial ratios, making many algorithms concrete. A well-based support uses a general well-order condition and is more flexible. Composition and fixed-point theorems require support hypotheses ensuring that substitution does not create infinitely many contributions above one monomial threshold [7].

The present ring avoids most of this machinery because its outer support is the integer ray $n \geq n_0$, and each inner log block is finite. Nevertheless, the same principles are visible in elementary form:

- finite decompositions give the Cauchy product;
- positive valuation gives summability;
- a contraction moves differences down the support;
- composition is valid when generator images have controlled support.

12.10 Analytic summability and resurgence

Polynomial-logarithmic transseries may be convergent, divergent but Poincaré-asymptotic, or part of a larger resurgent transseries with exponentially small sectors. The arithmetic in this article is indifferent to that distinction until one asks for an actual numerical value.

For a divergent expansion, one may need Borel transformation, analytic continuation in the Borel plane, Laplace integration, and Stokes data. Such operations lie beyond the present article, but the algebraic lesson remains: first organize the correct monomial hierarchy and formal equations; then apply an analytic summation theory compatible with it. Lambert's classical large-argument series is unusually benign because it is convergent on a substantial domain.

12.11 Choosing the smallest adequate algebra

There is a practical advantage to not starting with the largest possible transseries field. In a smaller algebra:

1. support comparison is cheaper;
2. closure failures identify genuinely new scales;
3. coefficient domains remain exact and understandable;
4. proofs can use elementary filtrations rather than general Hahn-series machinery.

A good symbolic system should therefore support a hierarchy of algebras and promote an expression only when an operation demands it:

$$\mathbb{K}[\ell]((t)) \subset \mathbb{K}((\ell^{-1}))((t)) \subset \text{Puisseux/Hahn power-log series} \subset \text{logarithmic-exponential transseries}.$$

Scale promotion

The correct extension is diagnosed by the first missing monomial. Division by ℓ asks for inverse log powers; algebraic roots ask for fractional outer powers; a leading ℓ^σ inside a logarithm asks for $\log \ell$; and composition with e^X asks for exponential height. Promotion should be explicit, not hidden inside simplification.

Chapter 13

Formula sheet and diagnostic checklist

13.1 Basic coordinates

$$t = X^{-1}, \quad \ell = \log X, \quad \mathcal{R} = \mathbb{K}[\ell][[t]], \quad \mathcal{P} = \mathbb{K}[\ell](t). \quad (13.1)$$

For $A = \sum P_n(\ell)t^n$,

$$v_t(A) = \min\{n : P_n \neq 0\}.$$

13.2 Differentiation

$$D_X = t\partial_\ell - t^2\partial_t, \quad (13.2)$$

$$D_X(t^n P) = t^{n+1}(P' - nP), \quad (13.3)$$

$$D_X^r(t^n P) = t^{n+r} \prod_{j=0}^{r-1} (D - n - j)P. \quad (13.4)$$

13.3 Multiplication and division

$$[t^n](AB) = \sum_{j+k=n} P_j Q_k, \quad (13.5)$$

$$C_n = b^{-1} \left(A_n - \sum_{j=1}^n B_j C_{n-j} \right) \quad (B_0 = b \in \mathbb{K}^\times), \quad (13.6)$$

$$V_{\text{new}} = V(2 - BV). \quad (13.7)$$

A Laurent series in t is a unit over $\mathbb{K}[\ell]$ exactly when its leading coefficient is a nonzero scalar.

13.4 Near-identity composition

For $G = X + A$, $A \in \mathcal{R}$,

$$t \circ G = \frac{t}{1 + tA}, \quad (13.8)$$

$$\ell \circ G = \ell + \log(1 + tA), \quad (13.9)$$

$$F \circ G = \sum_{j \geq 0} \frac{A^j}{j!} D_X^j F. \quad (13.10)$$

13.5 Series reversal

For $F = X + \phi$,

$$\psi = -\phi \circ (X + \psi), \quad F^{[-1]} = X + \psi, \quad (13.11)$$

$$H_{\text{new}} = H - \frac{F \circ H - X}{F' \circ H}, \quad (13.12)$$

$$F^{[-1]} = X + \sum_{m \geq 1} \frac{(-1)^m}{m!} D_X^{m-1}(\phi^m). \quad (13.13)$$

For $\phi = P(\ell)$,

$$[t^n](F^{[-1]} - X + P) = \frac{(-1)^{n+1}}{(n+1)!} D(D-1) \cdots (D-n+1) P^{n+1}. \quad (13.14)$$

13.6 Lambert polynomials

$$P_n(u) = \frac{(-1)^{n+1}}{(n+1)!} D(D-1) \cdots (D-n+1) u^{n+1}, \quad (13.15)$$

$$= \sum_{m=1}^n (-1)^{n-m} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] \frac{u^m}{m!}, \quad (13.16)$$

$$P_{n+1}(u) = -P_n(u) + n \int_0^u P_n(v) dv. \quad (13.17)$$

13.7 Lambert W

$$\xi_k = \text{Log } z + 2\pi i k, \quad (13.18)$$

$$W_k(z) \sim \xi_k - \log \xi_k + \sum_{n \geq 1} \frac{P_n(\log \xi_k)}{\xi_k^n}, \quad (13.19)$$

$$W_{-1}(x) \sim -\eta - \log \eta + \sum_{n \geq 1} (-1)^n \frac{P_n(\log \eta)}{\eta^n}, \quad \eta = \log(-1/x). \quad (13.20)$$

13.8 Diagnostic checklist

Before accepting a formal calculation, ask:

1. What is the asymptotic base point and sector?
2. What are the ordered monomials, and which one defines the outer valuation?
3. Is every infinite sum formally summable at the requested cutoff?
4. Does a denominator have an invertible leading coefficient in the declared coefficient ring?
5. Does composition send the generators into the same algebra?
6. Has a new scale such as ℓ^{-1} , $t^{1/q}$, $\log \ell$, or e^{-X} appeared?
7. Are branch choices consistent?
8. Was every intermediate truncated to a proven precision?
9. Does the final result satisfy an independent residual identity?

The governing pattern

A transseries operation is legitimate when every requested leading segment is determined by finitely much input data. In the polynomial-logarithmic ring, that principle is encoded by the t -adic valuation. Multiplication adds valuations, differentiation raises them, small-series functions push powers to higher order, and near-identity substitution contracts differences.

Part VI

Proofs, tables, and exercises

Appendix A

A formal proof of Lagrange–Bürmann inversion at infinity

This appendix supplies the algebra behind [Theorem 7.7](#). The ordinary Lagrange–Bürmann theorem is first proved with an auxiliary parameter. We then explain why the parameter can be specialized to one in the polynomial-logarithmic filtration, a step that is not legitimate in an arbitrary formal power-series ring. Standard general treatments of Lagrange inversion include [1, 8]; the proof below is adapted to the specific at-infinity filtration needed here.

A.1 Formal residue notation

Let C be a commutative $\mathbb{Q}$ -algebra. For a Laurent series

$$A(z) = \sum_{j \in \mathbb{Z}} a_j z^j \in C((z)),$$

define

$$\operatorname{Res}_z A(z) \, dz = a_{-1}.$$

Formal residues obey

$$\operatorname{Res}_z \frac{d}{dz} B(z) \, dz = 0 \tag{A.1}$$

for every Laurent series B . They are merely coefficient-extraction operators; no contour or analytic convergence is involved.

A.2 The parameter form of Lagrange–Bürmann

Consider

$$u = \varepsilon \Phi(u), \quad \Phi(z) \in C[[z]]. \tag{A.2}$$

Successive comparison of powers of ε gives a unique solution $u \in \varepsilon C[[\varepsilon]]$.

Theorem A.1 (Formal Lagrange–Bürmann formula). *For $H(z) \in C[[z]]$ and $m \geq 1$,*

$$[\varepsilon^m] H(u) = \frac{1}{m} [z^{m-1}] H'(z) \Phi(z)^m. \tag{A.3}$$

In particular,

$$[\varepsilon^m] u = \frac{1}{m} [z^{m-1}] \Phi(z)^m. \tag{A.4}$$

Proof. First suppose that $\Phi(0)$ is invertible. Then $\varepsilon = z/\Phi(z)$ has a compositional inverse, namely $z = u(\varepsilon)$, and formal change of variables for residues gives

$$\begin{aligned} [\varepsilon^m]H(u(\varepsilon)) &= \text{Res}_\varepsilon H(u(\varepsilon))\varepsilon^{-m-1} d\varepsilon \\ &= \text{Res}_z H(z)z^{-m-1}\Phi(z)^{m-1}(\Phi(z) - z\Phi'(z)) dz. \end{aligned}$$

By applying Equation (A.1) to $H(z)\Phi(z)^m z^{-m}$, one obtains

$$\begin{aligned} 0 &= \text{Res}_z \left(H'(z)\Phi(z)^m z^{-m} + mH(z)\Phi(z)^{m-1}\Phi'(z)z^{-m} \right. \\ &\quad \left. - mH(z)\Phi(z)^m z^{-m-1} \right) dz. \end{aligned}$$

Rearranging proves Equation (A.3). Taking $H(z) = z$ gives Equation (A.4).

For a general $\Phi(0)$, perform the calculation first in the universal polynomial ring generated by the coefficients of Φ , with the constant coefficient inverted. Both sides of Equation (A.3), at each fixed m , are polynomials with rational coefficients in only finitely many of those generators. Equality after localization is therefore a polynomial identity and survives every specialization, including specializations for which $\Phi(0)$ is not a unit. This removes the temporary invertibility assumption. $\square$

The more familiar differential form follows by Taylor expansion. If $\Phi(z) = \Psi(Y + z)$, then

$$[z^{m-1}]\Phi(z)^m = \frac{1}{(m-1)!} D_Y^{m-1}(\Psi(Y)^m).$$

Consequently, the solution of

$$u = \varepsilon\Psi(Y + u) \tag{A.5}$$

obeys

$$u = \sum_{m \geq 1} \frac{\varepsilon^m}{m!} D_Y^{m-1}(\Psi(Y)^m). \tag{A.6}$$

A.3 Specialization of the bookkeeping parameter

Set $\Psi = -\phi$, where $\phi \in \mathcal{R}$. Equation (A.5) becomes the inverse equation

$$u = -\varepsilon\phi(Y + u)$$

for $Y + \varepsilon\phi(Y)$. Formula (A.6) reads

$$u = \sum_{m \geq 1} \frac{(-\varepsilon)^m}{m!} D_Y^{m-1}(\phi(Y)^m). \tag{A.7}$$

A formal series in ε normally has no value at $\varepsilon = 1$. The present series does because differentiation raises the outer order:

$$\nu_t \left(D_Y^{m-1}(\phi^m) \right) \geq m\nu_t(\phi) + m - 1 \geq m - 1. \tag{A.8}$$

Thus, below any fixed t -order, only finitely many m contribute. The formal summability criterion Theorem 2.3 permits the coefficientwise specialization $\varepsilon = 1$, giving

$$(X + \phi)^{[-1]} = X + \sum_{m \geq 1} \frac{(-1)^m}{m!} D_X^{m-1}(\phi^m).$$

This also explains the exact finite-work rule: modulo t^N , terms with $m > N$ may be discarded; to extract the coefficient of t^N , terms with $m > N + 1$ may be discarded.

A.4 A transformed-observable version

The same residue calculation gives more than the inverse itself.

Theorem A.2 (Lagrange–Bürmann transport formula). *Let $F = X + \phi$, with $\phi \in \mathcal{R}$, and let $H \in \mathcal{P}$. Whenever both sides are interpreted to a fixed finite t -precision,*

$$H \circ F^{[-1]} = H + \sum_{m \geq 1} \frac{(-1)^m}{m!} D_X^{m-1}(H' \phi^m). \quad (\text{A.9})$$

Proof. Apply Equation (A.3) to the function $z \mapsto H(Y + z)$, with $\Phi(z) = -\phi(Y + z)$. Taylor coefficient extraction turns $[z^{m-1}]$ into $D_Y^{m-1}/(m-1)!$. Formal summability follows from

$$\nu_t \left(D_X^{m-1}(H' \phi^m) \right) \geq \nu_t(H) + m\nu_t(\phi) + m,$$

up to harmless improvement when derivatives vanish. Hence $\varepsilon = 1$ may be specialized coefficientwise. $\square$

Taking $H = X$ reduces Equation (A.9) to the inverse formula. Taking $H = \log X$, $H = X^{-q}$, or a known transseries observable gives its expansion along the inverse map without first constructing the entire inverse and then composing.

A.5 Pure polynomial-logarithmic perturbations revisited

For $\phi = P(\ell)$, the term indexed by m in the inverse formula is

$$\frac{(-1)^m}{m!} t^{m-1} D(D-1) \cdots (D-m+2) P(\ell)^m. \quad (\text{A.10})$$

It occupies exactly one outer block, t^{m-1} . Therefore the inverse is not merely computable order by order: every block has the closed expression

$$[t^n] \left((X + P(\ell))^{[-1]} - X + P(\ell) \right) = \frac{(-1)^{n+1}}{(n+1)!} D(D-1) \cdots (D-n+1) P(\ell)^{n+1}.$$

The Lambert polynomials are the special case $P(\ell) = \ell$.

Appendix B

Additional Lambert polynomials and coefficient diagonals

B.1 Polynomials through order twelve

The first eight polynomials were listed in [Equation \(8.16\)](#). Continuing the recurrence [Equation \(8.10\)](#) gives

$$\begin{aligned} P_9(u) &= \frac{u^9}{9} - \frac{761}{280}u^8 + \frac{29531}{1260}u^7 - \frac{1869}{20}u^6 + \frac{7483}{40}u^5 \\ &\quad - 189u^4 + 91u^3 - 18u^2 + u, \\ P_{10}(u) &= \frac{u^{10}}{10} - \frac{7129}{2520}u^9 + \frac{6515}{224}u^8 - \frac{9046}{63}u^7 + \frac{5985}{16}u^6 \\ &\quad - \frac{21091}{40}u^5 + \frac{1575}{4}u^4 - 145u^3 + \frac{45}{2}u^2 - u, \\ P_{11}(u) &= \frac{u^{11}}{11} - \frac{7381}{2520}u^{10} + \frac{177133}{5040}u^9 - \frac{420475}{2016}u^8 + \frac{341693}{504}u^7 \\ &\quad - \frac{60137}{48}u^6 + \frac{52591}{40}u^5 - \frac{3025}{4}u^4 + 220u^3 - \frac{55}{2}u^2 + u, \\ P_{12}(u) &= \frac{u^{12}}{12} - \frac{83711}{27720}u^{11} + \frac{2096083}{50400}u^{10} - \frac{3759217}{12960}u^9 + \frac{1533191}{1344}u^8 \\ &\quad - \frac{2667907}{1008}u^7 + \frac{146531}{40}u^6 - \frac{119141}{40}u^5 + \frac{5445}{4}u^4 - \frac{1925}{6}u^3 + 33u^2 - u. \end{aligned}$$

Each row can be checked either from the Stirling formula [Equation \(8.7\)](#) or from $P_{n+1} = -P_n + n \int_0^u P_n$.

B.2 General coefficient diagonals

Write

$$P_n(u) = \sum_{m=1}^n p_{n,m} u^m.$$

Then [Equation \(8.7\)](#) is equivalently

$$p_{n,m} = (-1)^{n-m} \frac{1}{m!} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right]. \quad (\text{B.1})$$

This formula exposes both the upper and lower coefficient diagonals. For fixed $r \geq 0$,

$$[u^{n-r}]P_n(u) = (-1)^r \frac{1}{(n-r)!} \left[\begin{matrix} n \\ r+1 \end{matrix} \right]. \quad (\text{B.2})$$

The first four are

$$[u^n]P_n = \frac{1}{n}, \quad (\text{B.3})$$

$$[u^{n-1}]P_n = -H_{n-1}, \quad (\text{B.4})$$

$$[u^{n-2}]P_n = \frac{n-1}{2} \left(H_{n-1}^2 - H_{n-1}^{(2)} \right), \quad (\text{B.5})$$

$$[u^{n-3}]P_n = -\frac{(n-1)(n-2)}{6} \left(H_{n-1}^3 - 3H_{n-1}H_{n-1}^{(2)} + 2H_{n-1}^{(3)} \right), \quad (\text{B.6})$$

where $H_N^{(q)} = \sum_{j=1}^N j^{-q}$. These follow from the classical harmonic-number expressions for $\left[\begin{matrix} n \\ 2 \end{matrix} \right]$, $\left[\begin{matrix} n \\ 3 \end{matrix} \right]$, and $\left[\begin{matrix} n \\ 4 \end{matrix} \right]$.

At the bottom of the coefficient triangle,

$$[u]P_n = (-1)^{n-1}, \quad (\text{B.7})$$

$$[u^2]P_n = (-1)^{n-2} \frac{n(n-1)}{4}, \quad (\text{B.8})$$

$$[u^3]P_n = (-1)^{n-3} \frac{n(n-1)(n-2)(3n-1)}{144}. \quad (\text{B.9})$$

For the last identity, use

$$\left[\begin{matrix} n \\ n-2 \end{matrix} \right] = 2 \binom{n}{3} + 3 \binom{n}{4} = \frac{n(n-1)(n-2)(3n-1)}{24}.$$

B.3 A coefficient recurrence

The differential recurrence [Equation \(8.8\)](#) gives a convenient recursion for individual coefficients. Equating the coefficient of u^{m-1} in

$$P'_{n+1} = nP_n - P'_n$$

yields

$$p_{n+1,m} = \frac{n}{m} p_{n,m-1} - p_{n,m}, \quad 1 \leq m \leq n+1, \quad (\text{B.10})$$

with the conventions $p_{n,0} = p_{n,n+1} = 0$ and $p_{1,1} = 1$. This is the polynomial recurrence stripped down to scalar rational arithmetic. It also makes the sign alternation immediate after multiplying by $(-1)^{n-m}$.

B.4 Exact arithmetic and denominator control

Formula [Equation \(B.1\)](#) shows that every denominator divides $m!$, and hence divides $n!$. Thus

$$n!P_n(u) \in \mathbb{Z}[u].$$

For exact symbolic computation, it is often advantageous to store the scaled polynomials

$$\widehat{P}_n(u) = n!P_n(u).$$

Their coefficients are integers, and the recurrence becomes

$$\widehat{P}_{n+1} = -(n+1)\widehat{P}_n + n(n+1) \int_0^u \widehat{P}_n(v) dv. \quad (\text{B.11})$$

The integral still divides each monomial coefficient by its new exponent, but integrality of the final result is guaranteed by the Stirling formula and can serve as an implementation invariant.

Appendix C

Exercises with solutions

The exercises are designed to test the algebra rather than numerical endurance. Every calculation should be truncated explicitly; silently manipulating infinite expressions is precisely the habit the filtered formalism is meant to replace.

C.1 Problems

Exercise C.1 (Cauchy multiplication). *Let*

$$A = 1 + (\ell + 1)t + (\ell^2 - 1)t^2, \quad B = 1 + (2 - \ell)t + \ell t^2.$$

Compute AB through order t^2 .

Exercise C.2 (Reciprocal of a unit). *Find C through order t^3 such that*

$$(1 + (\ell - 1)t + \ell t^2)C = 1 + O_t(t^4).$$

Explain why the reciprocal remains in $\mathcal{R}$.

Exercise C.3 (One-block integration). *Find a polynomial-logarithmic antiderivative of $t^2(\ell + 1)$. Verify it by applying $D_X = t\partial_\ell - t^2\partial_t$.*

Exercise C.4 (Composition with a constant translation). *For $G = X + a$, compute $(t\ell) \circ G$ through order t^3 , where $a \in \mathbb{K}$.*

Exercise C.5 (Taylor precision). *Let $F \in \mathcal{P}$ have $v_t(F) = s$, and let $A \in t^r \mathcal{R}$, $r \geq 0$. Prove that to compute $F \circ (X + A)$ modulo t^N , it suffices to retain Taylor indices*

$$0 \leq j \leq \left\lfloor \frac{N - 1 - s}{r + 1} \right\rfloor.$$

Exercise C.6 (Universal first inverse blocks). *Let $F = X + P(\ell)$, with $P \in \mathbb{K}[\ell]$. Express the inverse through order t^2 only in terms of P, P' , and P'' .*

Exercise C.7 (Quadratic logarithmic perturbation). *Reverse $F = X + \ell^2$ through order t^2 , and verify the degree law of [Theorem 7.9](#) in these first two nontrivial blocks.*

Exercise C.8 (A perturbation containing t). *Reverse $F = X + \ell + at$ through order t^2 .*

Exercise C.9 (Generating Lambert polynomials). Use $P_{n+1} = -P_n + n \int_0^u P_n(v) dv$ to generate P_2, P_3, P_4 from $P_1 = u$.

Exercise C.10 (Derivative of the Lambert expansion). Differentiate the large principal-branch expansion through L_1^{-3} and show that

$$W'(z) = \frac{1}{z} \left(1 - \frac{1}{L_1} + \frac{1 - L_2}{L_1^2} + \frac{-L_2^2 + 3L_2 - 1}{L_1^3} + O\left(\frac{L_2^3}{L_1^4}\right) \right).$$

Exercise C.11 (The lower real branch). Starting from $W_{-1}(x)e^{W_{-1}(x)} = x$, set $x = -e^{-\eta}$ and derive the first three correction blocks in [Equation \(9.17\)](#).

Exercise C.12 (Diagnosing scale promotion). For each expression below, name the smallest natural extension of $\mathbb{K}[\ell](t)$ required to represent it formally:

$$\frac{1}{\ell + t}, \quad X^{1/2}, \quad \log(X \log X), \quad e^{-X}.$$

Exercise C.13 (Transport along an inverse). Let $F = X + \phi$, $\phi \in \mathcal{R}$. Use [Theorem A.2](#) to write $(\log X) \circ F^{[-1]}$ through the terms indexed by $m = 3$. Do not first compute $F^{[-1]}$.

C.2 Solutions

Solution 1

The coefficient blocks are ordinary Cauchy products:

$$\begin{aligned} [t^0](AB) &= 1, \\ [t](AB) &= (\ell + 1) + (2 - \ell) = 3, \\ [t^2](AB) &= (\ell^2 - 1) + \ell + (\ell + 1)(2 - \ell) = 2\ell + 1. \end{aligned}$$

Hence

$$AB = 1 + 3t + (2\ell + 1)t^2 + O_t(t^3).$$

Solution 2

Write $C = 1 + c_1 t + c_2 t^2 + c_3 t^3 + O_t(t^4)$. Triangular coefficient comparison gives

$$\begin{aligned} c_1 &= 1 - \ell, \\ c_2 &= (\ell - 1)^2 - \ell = \ell^2 - 3\ell + 1, \\ c_3 &= -(\ell - 1)c_2 - \ell c_1 = -\ell^3 + 5\ell^2 - 5\ell + 1. \end{aligned}$$

Therefore

$$C = 1 + (1 - \ell)t + (\ell^2 - 3\ell + 1)t^2 + (-\ell^3 + 5\ell^2 - 5\ell + 1)t^3 + O_t(t^4).$$

The leading coefficient of the divisor is the unit $1 \in \mathbb{K}$, so [Theorem 5.1](#) guarantees a reciprocal in $\mathcal{R}$.

Solution 3

Seek an antiderivative in the form $tQ(\ell)$. Since

$$D_X(tQ) = t^2(Q' - Q),$$

we need $Q' - Q = \ell + 1$. The polynomial solution is $Q = -\ell - 2$, and hence

$$\int t^2(\ell + 1) dX = -t(\ell + 2) + C.$$

Indeed, $D_X[-t(\ell + 2)] = t^2(\ell + 1)$.

Solution 4

The generator substitutions are

$$\begin{aligned} t \circ G &= \frac{t}{1+at} = t - at^2 + a^2t^3 + O_t(t^4), \\ \ell \circ G &= \ell + \log(1+at) = \ell + at - \frac{a^2}{2}t^2 + \frac{a^3}{3}t^3 + O_t(t^4). \end{aligned}$$

Multiplying gives

$$(t\ell) \circ (X+a) = t\ell + a(1-\ell)t^2 + a^2\left(\ell - \frac{3}{2}\right)t^3 + O_t(t^4).$$

Solution 5

Differentiation raises outer order by at least one, so

$$\nu_t(D_X^j F) \geq s + j.$$

Also $\nu_t(A^j) \geq jr$. Thus the j -th Taylor term has order at least

$$s + j + jr = s + j(r+1).$$

It cannot affect the truncation below t^N once this quantity is at least N . The stated upper bound is exactly the largest integer j satisfying $s + j(r+1) < N$.

Solution 6

The Lagrange formula gives

$$F^{[-1]} = X - P + Q_1t + Q_2t^2 + O_t(t^3),$$

where

$$Q_1 = \frac{1}{2}D(P^2) = PP'$$

and

$$\begin{aligned} Q_2 &= -\frac{1}{6}D(D-1)(P^3) \\ &= \frac{1}{2}P^2P' - P(P')^2 - \frac{1}{2}P^2P''. \end{aligned}$$

Therefore

$$F^{[-1]} = X - P + PP't + \left(\frac{1}{2}P^2P' - P(P')^2 - \frac{1}{2}P^2P'' \right) t^2 + O_t(t^3).$$

Solution 7

Set $P = \ell^2$ in the preceding answer:

$$(X + \ell^2)^{[-1]} = X - \ell^2 + 2\ell^3 t + (\ell^5 - 5\ell^4)t^2 + O_t(t^3).$$

For $d = 2$, [Theorem 7.9](#) predicts degrees $2(n+1) - 1$, namely 3 and 5, and leading coefficients $2/n$, namely 2 and 1. Both are visible above.

Solution 8

Substitution in the fixed-point equation, or direct use of [Equation \(7.12\)](#), gives

$$(X + \ell + at)^{[-1]} = X - \ell + (\ell - a)t + \left(\frac{\ell^2}{2} - \ell - a\ell + a\right)t^2 + O_t(t^3).$$

Composing this truncation with $X + \ell + at$ leaves a residual $O_t(t^3)$.

Solution 9

Successive integration gives

$$\begin{aligned} P_2 &= -u + \int_0^u v \, dv = \frac{u^2}{2} - u, \\ P_3 &= -P_2 + 2 \int_0^u P_2(v) \, dv = \frac{u^3}{3} - \frac{3}{2}u^2 + u, \\ P_4 &= -P_3 + 3 \int_0^u P_3(v) \, dv = \frac{u^4}{4} - \frac{11}{6}u^3 + 3u^2 - u. \end{aligned}$$

Solution 10

Because $L_1 = \log z$,

$$\frac{d}{dz} = \frac{1}{z} \frac{d}{dL_1}, \quad \frac{dL_2}{dL_1} = \frac{1}{L_1}.$$

Differentiate

$$W = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2^2/2 - L_2}{L_1^2} + \frac{L_2^3/3 - 3L_2^2/2 + L_2}{L_1^3} + \dots$$

blockwise. Terms displayed in the last numerator begin affecting the derivative only at order L_1^{-4} . Collecting through L_1^{-3} gives exactly the expression in the problem. The same result follows by expanding $W/[z(1+W)]$.

Solution 11

Put $W_{-1}(x) = -v$. Since $x = -e^{-\eta}$, the defining equation is

$$ve^{-v} = e^{-\eta}, \quad v - \log v = \eta.$$

Equivalently, substitute $L_1 = -\eta$ into the universal large-branch Lambert expansion while keeping $L = \log \eta$ real. The first blocks are

$$W_{-1}(x) = -\eta - L - \frac{L}{\eta} + \frac{L^2/2 - L}{\eta^2} - \frac{L^3/3 - 3L^2/2 + L}{\eta^3} + O\left(\frac{L^4}{\eta^4}\right).$$

Solution 12

The reciprocal of $\ell + t$ requires inverse powers of ℓ , so a natural home is $\mathbb{K}((\ell^{-1}))((t))$. The monomial $X^{1/2} = t^{-1/2}$ requires a Puiseux extension. The identity

$$\log(X \log X) = \ell + \log \ell$$

requires a second logarithmic level. Finally, e^{-X} is beyond every power-log scale and requires an exponential transmonomial, hence a logarithmic-exponential transseries extension.

Solution 13

Take $H = \log X$, so $H' = t$. Formula [Equation \(A.9\)](#) gives

$$(\log X) \circ F^{[-1]} = \ell - t\phi + \frac{1}{2}D_X(t\phi^2) - \frac{1}{6}D_X^2(t\phi^3) + \text{terms with } m \geq 4.$$

One may simplify the derivatives with $D_X(t^n P) = t^{n+1}(P' - nP)$ whenever ϕ is given in block form. The expression is obtained directly along the inverse map; no separate construction of $F^{[-1]}$ was used.

Appendix D

Historical and bibliographic notes

D.1 From logarithmic-exponential germs to transseries

Hardy's study of orders of infinity organized nonoscillatory elementary germs by eventual growth [9]. Later work on exponential-logarithmic terms and generalized power series developed increasingly algebraic versions of this hierarchy [4, 14, 15]. Modern transseries fields combine ordered supports, field operations, exponential and logarithmic extensions, derivation, integration, and composition. Van der Hoeven's monograph gives a systematic differential-algebraic and algorithmic treatment [16]; Edgar's expository article and composition notes give an accessible entry to grid-based and well-based viewpoints [6, 7].

The present article deliberately works in a much smaller algebra. The restriction to integer powers of X^{-1} and polynomial coefficients in $\log X$ turns general support questions into ordinary coefficient recursions. The resulting theory is not a competing definition of full transseries; it is a computationally useful local chart inside them.

D.2 Power-logarithmic terminology

The phrases *power-log series*, *logarithmic transseries*, and *Dulac series* are used with somewhat different support conventions in different subjects. In local dynamical systems, power-log transseries often use real powers of a small variable and integer powers of its logarithm. The algebras studied by Mardešić, Resman, Rolin, and Županović are important examples [10]. Under $x = X^{-1}$, their monomials $x^\alpha (-\log x)^k$ become $X^{-\alpha} (\log X)^k$, which explains the close relation to the present large-variable notation.

Here *polynomial-logarithmic* means specifically that each fixed outer block has a coefficient in $\mathbb{K}[\log X]$. The word *polynomial* does not restrict the infinitely many outer blocks, and it does not assert closure under arbitrary division.

D.3 Symbolic asymptotics and inverse functions

Algorithms for asymptotic scales, implicit functions, and inverse expansions were developed in the symbolic-asymptotics literature. The work of Salvy and Shackell is especially relevant to multiseries and functional inversion [11, 12, 13]. The filtered algorithms in this article are specialized versions adapted to one outer power scale and one logarithmic coefficient scale. Their advantage is that precision propagation, closure, and failure modes can all be stated in a few valuation inequalities.

D.4 Lambert W sources

The modern reference on Lambert W , including branches, asymptotics, and computation, is the paper of Corless, Gonnet, Hare, Jeffrey, and Knuth [2]. Corless, Jeffrey, and Knuth developed related sequences of convergent series and numerical approximants [3]. The NIST Digital Library of Mathematical Functions records the standard large-branch expansions and their domains [5]. The main contribution of the present treatment is organizational: the Lambert coefficients are placed inside a general arithmetic, composition, and reversion calculus, and their Stirling formula is derived as a direct specialization of the at-infinity Lagrange–Bürmann operator.

D.5 Reversion versus reciprocal inversion

Older literature often uses *reversion* for compositional inversion of a series, while *inverse* may mean either a multiplicative reciprocal or a compositional inverse. This article uses A^{-1} for reciprocals and $F^{[-1]}$ for compositional inverses. The phrase *series reversal* in the title is synonymous with reversion; it emphasizes reversal of the map rather than inversion of its values.

Bibliography

- [1] L. Comtet, *Advanced Combinatorics: The Art of Finite and Infinite Expansions*, D. Reidel, Dordrecht, 1974.
- [2] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, “On the Lambert W function,” *Advances in Computational Mathematics* **5** (1996), 329–359. doi:10.1007/BF02124750.
- [3] R. M. Corless, D. J. Jeffrey, and D. E. Knuth, “A sequence of series for the Lambert W function,” in *Proceedings of ISSAC '97*, ACM, 1997, pp. 197–204. doi:10.1145/258726.258783.
- [4] B. I. Dahn and P. Göring, “Notes on exponential-logarithmic terms,” *Fundamenta Mathematicae* **127** (1987), no. 1, 45–50. doi:10.4064/fm-127-1-45-50.
- [5] NIST Digital Library of Mathematical Functions, §4.13, “Lambert W -Function,” F. W. J. Olver et al., eds., National Institute of Standards and Technology. <https://dlmf.nist.gov/4.13>.
- [6] G. A. Edgar, “Transseries for beginners,” *Real Analysis Exchange* **35** (2010), no. 2, 253–310. doi:10.14321/realanalexch.35.2.0253; arXiv:0801.4877.
- [7] G. A. Edgar, “Transseries: composition, recursion, and convergence,” 2009. arXiv:0909.1259.
- [8] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, Cambridge, 2009.
- [9] G. H. Hardy, *Orders of Infinity*, Cambridge Tracts in Mathematics and Mathematical Physics, no. 12, Cambridge University Press, Cambridge, 1910.
- [10] P. Mardešić, M. Resman, J.-P. Rolin, and V. Županović, “Normal forms and embeddings for power-log transseries,” *Advances in Mathematics* **303** (2016), 888–953. doi:10.1016/j.aim.2016.08.030.
- [11] B. Salvy and J. Shackell, “Asymptotic expansions of functional inverses,” in P. S. Wang, ed., *Symbolic and Algebraic Computation: Proceedings of ISSAC '92*, ACM Press, 1992, pp. 130–137.
- [12] B. Salvy and J. Shackell, “Symbolic asymptotics: functions of two variables, implicit functions,” *Journal of Symbolic Computation* **25** (1998), no. 3, 329–349. doi:10.1006/jsco.1997.0179.
- [13] B. Salvy and J. Shackell, “Symbolic asymptotics: multiseries of inverse functions,” *Journal of Symbolic Computation* **27** (1999), no. 6, 543–563. doi:10.1006/jsco.1999.0281.
- [14] L. van den Dries, A. Macintyre, and D. Marker, “Logarithmic-exponential power series,” *Journal of the London Mathematical Society* (2) **56** (1997), no. 3, 417–434. doi:10.1112/S0024610797005437.
- [15] L. van den Dries, A. Macintyre, and D. Marker, “Logarithmic-exponential series,” *Annals of Pure and Applied Logic* **111** (2001), nos. 1–2, 61–113. doi:10.1016/S0168-0072(01)00035-5.

- [16] J. van der Hoeven, *Transseries and Real Differential Algebra*, Lecture Notes in Mathematics, vol. 1888, Springer, Berlin, 2006. [doi:10.1007/3-540-35590-1](https://doi.org/10.1007/3-540-35590-1).